

CHAPTER 2:

Quadratic Equations and Inequalities

EXERCISE 2.1

BOARD EXAM
PREPARATION SERIES

QUESTION 1

Write the following quadratic equations in standard form.

STANDARD FORM OF A QUADRATIC EQUATION: $ax^2 + bx + c = 0$ WHERE $a \neq 0$

KEY FACTS

QUADRATIC EQUATION

$$ax^2 + bx + c = 0$$

where $a \neq 0$

**DEGREE OF
QUADRATIC EQUATION**

Degree = 2

STANDARD FORM

$$ax^2 + bx + c = 0$$

IMPORTANT CONDITION

$a \neq 0$

NOTE

In standard form
the coefficient of
 x^2 must be
non-zero.

(i) $3x - 1 = 2x^2$

SOLUTION:

GIVEN:

$$3x - 1 = 2x^2$$

BRING ALL TERMS TO ONE SIDE:

$$3x - 1 - 2x^2 = 0$$

$$-(2x^2 - 3x + 1) = 0$$

$$0 = 2x^2 - 3x + 1$$

THEREFORE,

$$2x^2 - 3x + 1 = 0$$

FINAL ANSWER:

$$2x^2 - 3x + 1 = 0$$

(ii) $2x(x + 1) = 4(2x + 3)$

SOLUTION:

GIVEN:

$$2x(x + 1) = 4(2x + 3)$$

OPEN THE BRACKETS:

$$2x^2 + 2x = 8x + 12$$

BRING ALL TERMS TO THE LEFT SIDE:

$$2x^2 + 2x - 8x - 12 = 0$$

COMBINE LIKE TERMS:

$$2x^2 - 6x - 12 = 0$$

DIVIDE BY 2:

$$x^2 - 3x - 6 = 0$$

FINAL ANSWER:

$$x^2 - 3x - 6 = 0$$

(iii) $2x^2 - 4x = 4x + 7$

SOLUTION:

GIVEN:

$$2x^2 - 4x = 4x + 7$$

BRING ALL TERMS TO THE LEFT SIDE:

$$2x^2 - 4x - 4x - 7 = 0$$

COMBINE LIKE TERMS:

$$2x^2 - 8x - 7 = 0$$

FINAL ANSWER:

$$2x^2 - 8x - 7 = 0$$

(iv) $4(3x - 2) = 9x^2$

SOLUTION:

GIVEN:

$$4(3x - 2) = 9x^2$$

OPEN THE BRACKET:

$$12x - 8 = 9x^2$$

BRING ALL TERMS TO ONE SIDE:

$$12x - 8 - 9x^2 = 0$$

$$-(9x^2 - 12x + 8) = 0$$

$$0 = 9x^2 - 12x + 8$$

THEREFORE,

$$9x^2 - 12x + 8 = 0$$

FINAL ANSWER:

$$9x^2 - 12x + 8 = 0$$

(v) $2x + \frac{1}{x} = 5 - \frac{1}{x}, x \neq 0$

SOLUTION:

GIVEN:

$$2x + \frac{1}{x} = 5 - \frac{1}{x}$$

ADD $\frac{1}{x}$ TO BOTH SIDES:

$$2x + \frac{1}{x} + \frac{1}{x} = 5$$

$$2x + \frac{2}{x} = 5$$

MULTIPLY BOTH SIDES BY x :

$$x(2x + \frac{2}{x}) = 5x$$

$$2x^2 + 2 = 5x$$

BRING ALL TERMS TO ONE SIDE:

$$2x^2 + 2 - 5x = 0$$

$$2x^2 - 5x + 2 = 0$$

FINAL ANSWER:

$$2x^2 - 5x + 2 = 0$$

(vi) $\frac{6x+6}{20-x} = \frac{1}{x}, x \neq 0, 20$

SOLUTION:

GIVEN:

$$\frac{6x+6}{20-x} = \frac{1}{x}$$

CROSS MULTIPLY:

$$x(6x+6) = 20-x$$

OPEN THE BRACKET:

$$6x^2 + 6x = 20 - x$$

BRING ALL TERMS TO THE LEFT SIDE:

$$6x^2 + 6x + x - 20 = 0$$

$$6x^2 + 7x - 20 = 0$$

FINAL ANSWER:

$$6x^2 + 7x - 20 = 0$$

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QUESTION 2 Solve the following quadratic equations by factorization method.

(i) $x^2 - x - 6 = 0$

Solution:

Given:

$$x^2 - x - 6 = 0$$

Find two numbers whose:

$$\text{Product} = -6$$

$$\text{Sum} = -1$$

The numbers are: $-3, 2$

Split the middle term:

$$x^2 - 3x + 2x - 6 = 0$$

Take common factors:

$$x(x - 3) + 2(x - 3) = 0$$

$$(x - 3)(x + 2) = 0$$

Therefore,

$$x - 3 = 0 \quad \text{or} \quad x + 2 = 0$$

$$x = 3 \quad \text{or} \quad x = -2$$

FINAL ANSWER

$$x = 3, -2$$

(ii) $x^2 + 3x - 28 = 0$

Solution:

Given:

$$x^2 + 3x - 28 = 0$$

Find two numbers whose:

$$\text{Product} = -28$$

$$\text{Sum} = 3$$

The numbers are: $7, -4$

Split the middle term:

$$x^2 + 7x - 4x - 28 = 0$$

Take common factors:

$$x(x + 7) - 4(x + 7) = 0$$

$$(x + 7)(x - 4) = 0$$

Therefore,

$$x + 7 = 0 \quad \text{or} \quad x - 4 = 0$$

$$x = -7 \quad \text{or} \quad x = 4$$

FINAL ANSWER

$$x = 4, -7$$

(iii) $6x^2 + 13x - 5 = 0$

Solution:

Given:

$$6x^2 + 13x - 5 = 0$$

Multiply:

$$6 \times (-5) = -30$$

Find two numbers whose:

$$\text{Product} = -30$$

$$\text{Sum} = 13$$

The numbers are: $15, -2$

Split the middle term:

$$6x^2 + 15x - 2x - 5 = 0$$

Group the terms:

$$(6x^2 + 15x) + (-2x - 5) = 0$$

Take common factors:

$$3x(2x + 5) - 1(2x + 5) = 0$$

$$(2x + 5)(3x - 1) = 0$$

Therefore,

$$2x + 5 = 0 \quad \text{or} \quad 3x - 1 = 0$$

$$2x = -5 \quad 3x = 1$$

$$x = -\frac{5}{2}$$

$$x = \frac{1}{3}$$

FINAL ANSWER

$$x = -\frac{5}{2}, \frac{1}{3}$$

KEY FACTS

Factorization Method

1. Write equation in standard form.
2. Find two numbers whose product = ac and sum = b .
3. Split middle term.
4. Take common factors.
5. Apply Zero Product Rule.

Zero Product Rule

$$ab = 0$$

then

$$a = 0 \quad \text{or} \quad b = 0$$

Standard Form of Quadratic Equation

$$ax^2 + bx + c = 0$$

where $a \neq 0$

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QUESTION 2

Solve the following quadratic equations by factorization method.

(iv) $x^2 - \frac{3}{2}x = \frac{9}{2}$

SOLUTION:

Given:

$$x^2 - \frac{3}{2}x = \frac{9}{2}$$

Bring all terms to one side:

$$x^2 - \frac{3}{2}x - \frac{9}{2} = 0$$

Multiply the whole equation by 2 to remove fractions:

$$2x^2 - 3x - 9 = 0$$

Find two numbers whose:

$$\begin{aligned} \text{Product} &= 2 \times (-9) = -18 \\ \text{Sum} &= -3 \end{aligned}$$

The numbers are: -6, 3

Split the middle term:

$$2x^2 - 6x + 3x - 9 = 0$$

Group the terms:

$$(2x^2 - 6x) + (3x - 9) = 0$$

Take common factors:

$$2x(x - 3) + 3(x - 3) = 0$$

$$(x - 3)(2x + 3) = 0$$

Therefore,

$$x - 3 = 0 \quad \text{or} \quad 2x + 3 = 0$$

$$\begin{aligned} x &= 3 \quad \text{or} \quad 2x = -3 \\ x &= -\frac{3}{2} \end{aligned}$$

FINAL ANSWER:

$$x = 3, -\frac{3}{2}$$

(v) $\frac{3x-8}{x-2} = \frac{5x-2}{x+5}, x \neq 2, -5$

SOLUTION:

Given:

$$\frac{3x-8}{x-2} = \frac{5x-2}{x+5}, x \neq 2, -5$$

Cross multiply:

$$(3x-8)(x+5) = (5x-2)(x-2)$$

Expand both sides:

$$3x(x+5) - 8(x+5) = 5x(x-2) - 2(x-2)$$

$$3x^2 + 15x - 8x - 40 = 5x^2 - 10x - 2x + 4$$

$$3x^2 + 7x - 40 = 5x^2 - 12x + 4$$

Bring all terms to one side:

$$\begin{aligned} 3x^2 + 7x - 40 - 5x^2 + 12x - 4 &= 0 \\ -2x^2 + 19x - 44 &= 0 \end{aligned}$$

Multiply by -1:

$$2x^2 - 19x + 44 = 0$$

Find two numbers whose:

$$\begin{aligned} \text{Product} &= 2 \times 44 = 88 \\ \text{Sum} &= -19 \end{aligned}$$

The numbers are: -11, -8

Split the middle term:

$$2x^2 - 11x - 8x + 44 = 0$$

Group the terms:

$$(2x^2 - 11x) + (-8x + 44) = 0$$

Take common factors:

$$\begin{aligned} x(2x - 11) - 4(2x - 11) &= 0 \\ (2x - 11)(x - 4) &= 0 \end{aligned}$$

Therefore,

$$\begin{aligned} 2x - 11 &= 0 \quad \text{or} \quad x - 4 = 0 \\ 2x &= 11 \quad \text{or} \quad x = 4 \\ x &= \frac{11}{2} \end{aligned}$$

FINAL ANSWER:

$$\begin{aligned} x &= \frac{11}{2}, 4 \\ (\text{Check: } x \neq 2, -5 \quad \checkmark) \end{aligned}$$

(vi) $\frac{1}{x-1} - \frac{1}{x+3} = \frac{1}{35}, x \neq 1, -3$

SOLUTION:

Given:

$$\frac{1}{x-1} - \frac{1}{x+3} = \frac{1}{35}, x \neq 1, -3$$

Take L.C.M. of denominators:

$$\text{L.C.M.} = (x-1)(x+3)$$

Multiply both sides by $(x-1)(x+3)$:

$$(x+3) - (x-1) = \frac{1}{35}(x-1)(x+3)$$

Simplify:

$$x+3-x+1 = \frac{1}{35}(x^2+3x-x-3)$$

$$4 = \frac{1}{35}(x^2+2x-3)$$

Multiply both sides by 35:

$$140 = x^2 + 2x - 3$$

Bring all terms to one side:

$$x^2 + 2x - 3 - 140 = 0$$

$$x^2 + 2x - 143 = 0$$

Find two numbers whose:

$$\begin{aligned} \text{Product} &= -143 \\ \text{Sum} &= 2 \end{aligned}$$

The numbers are: 13, -11

Split the middle term:

$$x^2 + 13x - 11x - 143 = 0$$

Group the terms:

$$(x^2 + 13x) + (-11x - 143) = 0$$

Take common factors:

$$\begin{aligned} x(x+13) - 11(x+13) &= 0 \\ (x+13)(x-11) &= 0 \end{aligned}$$

Therefore,

$$\begin{aligned} x+13 &= 0 \quad \text{or} \quad x-11 = 0 \\ x &= -13 \quad \text{or} \quad x = 11 \end{aligned}$$

FINAL ANSWER:

$$\begin{aligned} x &= -13, 11 \\ (\text{Check: } x \neq 1, -3 \quad \checkmark) \end{aligned}$$

KEY FACTS

Factorization Method

1. Write equation in standard form.
2. Find two numbers whose product = ac and sum = b .
3. Split middle term.
4. Take common factors.
5. Apply Zero Product Rule.

Zero Product Rule

$$ab = 0$$

then

$$a = 0 \quad \text{or} \quad b = 0$$

Standard Form of Quadratic Equation

$$ax^2 + bx + c = 0$$

where: $a \neq 0$

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QUESTION 3

Solve the following quadratic equations by completing square method.

(i) $2x^2 + 5x + 2 = 0$

SOLUTION:

GIVEN:

$$2x^2 + 5x + 2 = 0$$

STEP 1: Make coefficient of x^2 equal to 1

$$2x^2 + 5x + 2 = 0$$

$$\Rightarrow x^2 + \frac{5}{2}x + 1 = 0$$

STEP 2: Move constant to RHS

$$x^2 + \frac{5}{2}x = -1$$

STEP 3: Add $\left(\frac{b}{2}\right)^2 = \left(\frac{5}{4}\right)^2 = \frac{25}{16}$

$$x^2 + \frac{5}{2}x + \frac{25}{16} = -1 + \frac{25}{16}$$

$$\left(x + \frac{5}{4}\right)^2 = \frac{9}{16}$$

STEP 4: Take square root

$$x + \frac{5}{4} = \pm \frac{3}{4}$$

STEP 5: Find x

$$x = -\frac{5}{4} + \frac{3}{4} = -\frac{2}{4} = -\frac{1}{2}$$

or

$$x = -\frac{5}{4} - \frac{3}{4} = -\frac{8}{4} = -2$$

THEREFORE,

$$x = -\frac{1}{2}, -2$$

FINAL ANSWER:

$$x = -\frac{1}{2}, -2$$

(ii) $x^2 + x = 42$

SOLUTION:

GIVEN:

$$x^2 + x = 42$$

STEP 1: Move constant to LHS

$$x^2 + x - 42 = 0$$

STEP 2: Add $\left(\frac{b}{2}\right)^2 = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$

$$x^2 + x + \frac{1}{4} = 42 + \frac{1}{4}$$

$$\left(x + \frac{1}{2}\right)^2 = \frac{169}{4}$$

STEP 3: Take square root

$$x + \frac{1}{2} = \pm \frac{13}{2}$$

STEP 4: Find x

$$x = -\frac{1}{2} + \frac{13}{2} = \frac{12}{2} = 6$$

or

$$x = -\frac{1}{2} - \frac{13}{2} = -\frac{14}{2} = -7$$

THEREFORE,

$$x = 6, -7$$

FINAL ANSWER:

$$x = 6, -7$$

KEY FACTS

COMPLETING SQUARE FORMULA

$$x^2 + bx + c = 0$$

$$\left(x + \frac{b}{2}\right)^2 = \frac{b^2}{4} - c$$

DEFINITION

Completing square method is a method used to solve quadratic equations by making the expression a perfect square.

GENERAL FORM

$$ax^2 + bx + c = 0, \\ a \neq 0$$

CONDITION

$$a \neq 0$$

EXAM TIP

Always take half of the coefficient of x , square it, and add on both sides.

SHORT NOTE

After getting square on one side, take square root on both sides to find the values of x .

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QUESTION 3

Solve the following quadratic equations by completing square method.

(iii) $12x^2 + 7x = 12$

SOLUTION:

GIVEN:

$$12x^2 + 7x = 12$$

STEP 1: Rearrange in standard form

$$12x^2 + 7x - 12 = 0$$

$$\Rightarrow 12x^2 + 7x = 12$$

STEP 2: Make coefficient of x^2 equal to 1

Divide both sides by 12:

$$x^2 + \frac{7}{12}x = 1$$

STEP 3: Add $\left(\frac{b}{2}\right)^2 = \left(\frac{7}{24}\right)^2 = \frac{49}{576}$ to both sides

$$x^2 + \frac{7}{12}x + \frac{49}{576} = 1 + \frac{49}{576}$$

$$\left(x + \frac{7}{24}\right)^2 = \frac{576 + 49}{576} = \frac{625}{576}$$

STEP 4: Take square root

$$x + \frac{7}{24} = \pm \frac{25}{24}$$

STEP 5: Find x

$$x = -\frac{7}{24} + \frac{25}{24} = \frac{18}{24} = \frac{3}{4}$$

or

$$x = -\frac{7}{24} - \frac{25}{24} = -\frac{32}{24} = -\frac{4}{3}$$

THEREFORE,

$$x = \frac{3}{4}, -\frac{4}{3}$$

FINAL ANSWER:

$$x = \frac{3}{4}, -\frac{4}{3}$$

(iv) $\frac{x+3}{2x-7} = \frac{2x-1}{x-3}, x \neq \frac{7}{2}, 3$

SOLUTION:

GIVEN:

$$\frac{x+3}{2x-7} = \frac{2x-1}{x-3}, x \neq \frac{7}{2}, 3$$

STEP 1: Cross multiply

$$(x+3)(x-3) = (2x-1)(2x-7)$$

STEP 2: Expand both sides

$$x^2 - 9 = 4x^2 - 14x - 2x + 7$$

$$x^2 - 9 = 4x^2 - 16x + 7$$

STEP 3: Bring all terms to one side

$$x^2 - 9 - 4x^2 + 16x - 7 = 0$$

$$-3x^2 + 16x - 16 = 0$$

$$\Rightarrow 3x^2 - 16x + 16 = 0$$

STEP 4: Make coefficient of x^2 equal to 1

Divide both sides by 3:

$$x^2 - \frac{16}{3}x + \frac{16}{3} = 0$$

$$\Rightarrow x^2 - \frac{16}{3}x = -\frac{16}{3}$$

STEP 5: Add $\left(\frac{b}{2}\right)^2 = \left(\frac{-16}{6}\right)^2 = \frac{64}{9}$ to both sides

$$x^2 - \frac{16}{3}x + \frac{64}{9} = -\frac{16}{3} + \frac{64}{9}$$

$$\left(x - \frac{8}{3}\right)^2 = \frac{-48 + 64}{9} = \frac{16}{9}$$

STEP 6: Take square root

$$x - \frac{8}{3} = \pm \frac{4}{3}$$

STEP 7: Find x

$$x = \frac{8}{3} + \frac{4}{3} = \frac{12}{3} = 4$$

or

$$x = \frac{8}{3} - \frac{4}{3} = \frac{4}{3}$$

Check: $x \neq \frac{7}{2}, 3$ (Both solutions are valid.)

THEREFORE,

$$x = 4, \frac{4}{3}$$

FINAL ANSWER:

$$x = 4, \frac{4}{3}$$

KEY FACTS

COMPLETING SQUARE FORMULA

$$x^2 + bx + c = 0$$

$$\left(x + \frac{b}{2}\right)^2 = \frac{b^2}{4} - c$$

DEFINITION

Completing square method is a method used to solve quadratic equations by making the expression a perfect square.

GENERAL FORM

$$ax^2 + bx + c = 0, \\ a \neq 0$$

CONDITION

$$a \neq 0$$

EXAM TIP

Always take half of the coefficient of x, square it, and add on both sides.

SHORT NOTE

After getting square on one side, take square root on both sides to find the values of x.

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QUESTION 3

Solve the following quadratic equations by completing square method.

(v) $\frac{1}{1+x} - \frac{1}{3-x} = \frac{6}{35}, x \neq -1, 3$

SOLUTION:

GIVEN:

$$\frac{1}{1+x} - \frac{1}{3-x} = \frac{6}{35}, x \neq -1, 3$$

STEP 1: Take LCM of denominators

$$\text{LCM of } (1+x) \text{ and } (3-x) = (1+x)(3-x)$$

Multiply both sides by $(1+x)(3-x)$:

$$(3-x) - (1+x) = \frac{6}{35} (1+x)(3-x)$$

STEP 2: Simplify

$$3-x-1-x = \frac{6}{35} (3-x^2-2x)$$

$$2-2x = \frac{6}{35} (3-2x-x^2)$$

STEP 3: Bring all terms to one side

$$35(2-2x) = 6(3-2x-x^2)$$

$$70-70x = 18-12x-6x^2$$

$$6x^2-12x-70x+18-70=0$$

$$6x^2-82x-52=0$$

Divide by 2:

$$3x^2-41x-26=0$$

STEP 4: Factorize

$$3x^2-41x-26=0$$

$$(3x-2)(x-13)=0$$

STEP 5: Find x

$$3x-2=0 \quad \text{or} \quad x-13=0$$

$$3x=2$$

$$x=13$$

$$x = \frac{2}{3}$$

$$x = 13$$

Check: $x \neq -1, 3$ (Both values are valid.)

THEREFORE,

$$x = \frac{2}{3}, 13$$

FINAL ANSWER:

$$x = \frac{2}{3}, 13$$

(vi) $\frac{3x-1}{4x+7} = 1 - \frac{6}{x+7}, x \neq -\frac{7}{4}, -7$

SOLUTION:

GIVEN:

$$\frac{3x-1}{4x+7} = 1 - \frac{6}{x+7}, x \neq -\frac{7}{4}, -7$$

STEP 1: Write RHS with common denominator

$$1 = \frac{x+7}{x+7}$$

So,

$$\frac{3x-1}{4x+7} = \frac{x+7}{x+7} - \frac{6}{x+7} = \frac{x+1}{x+7}$$

STEP 2: Cross multiply

$$(3x-1)(x+7) = (x+1)(4x+7)$$

STEP 3: Expand both sides

$$3x^2+21x-x-7 = 4x^2+7x+4x+7$$

$$3x^2+20x-7 = 4x^2+11x+7$$

STEP 4: Bring all terms to one side

$$3x^2+20x-7-4x^2-11x-7=0$$

$$-x^2+9x-14=0$$

Multiply by (-1):

$$x^2-9x+14=0$$

STEP 5: Factorize

$$x^2-9x+14=0$$

$$(x-7)(x-2)=0$$

STEP 6: Find x

$$x-7=0 \quad \text{or} \quad x-2=0$$

$$x=7$$

$$x=2$$

Check: $x \neq -\frac{7}{4}, -7$ (Both values are valid.)

THEREFORE,

$$x = 7, 2$$

FINAL ANSWER:

$$x = 7, 2$$

KEY FACTS

COMPLETING SQUARE FORMULA

$$x^2 + bx + c = 0$$

$$\left(x + \frac{b}{2}\right)^2 = \frac{b^2}{4} - c$$

DEFINITION

Completing square method is a method used to solve quadratic equations by making the expression a perfect square.

GENERAL FORM

$$ax^2 + bx + c = 0, \\ a \neq 0$$

CONDITION

$$a \neq 0$$

EXAM TIP

Always take half of the coefficient of x , square it, and add on both sides.

SHORT NOTE

After getting square on one side, take square root on both sides to find the values of x .

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QUESTION 4

Use quadratic formula to solve the following equations:

(i) $2x^2 - 5x + 3 = 0$

SOLUTION:

Given:

$$2x^2 - 5x + 3 = 0$$

Comparing with $ax^2 + bx + c = 0$, we get

$$a = 2, \quad b = -5, \quad c = 3$$

Using quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(-5) \pm \sqrt{(-5)^2 - 4(2)(3)}}{2(2)}$$

$$x = \frac{5 \pm \sqrt{25 - 24}}{4}$$

$$x = \frac{5 \pm \sqrt{1}}{4}$$

$$x = \frac{5 \pm 1}{4}$$

So,

$$x = \frac{5+1}{4} = \frac{6}{4} = \frac{3}{2} \quad x = \frac{5-1}{4} = \frac{4}{4} = 1$$

THEREFORE,

$$x = \frac{3}{2}, 1$$

(ii) $2x^2 - 7x - 15 = 0$

SOLUTION:

Given:

$$2x^2 - 7x - 15 = 0$$

Comparing with $ax^2 + bx + c = 0$, we get

$$a = 2, \quad b = -7, \quad c = -15$$

Using quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(-7) \pm \sqrt{(-7)^2 - 4(2)(-15)}}{2(2)}$$

$$x = \frac{7 \pm \sqrt{49 + 120}}{4}$$

$$x = \frac{7 \pm \sqrt{169}}{4}$$

$$x = \frac{7 \pm 13}{4}$$

So,

$$x = \frac{7+13}{4} = \frac{20}{4} = 5 \quad x = \frac{7-13}{4} = \frac{-6}{4} = -\frac{3}{2}$$

THEREFORE,

$$x = 5, -\frac{3}{2}$$

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QUESTION 4

Use quadratic formula to solve the following equations:

(iii) $2x^2 + 7x = 15$

SOLUTION:

Given:

$$2x^2 + 7x = 15$$

Rearranging,

$$2x^2 + 7x - 15 = 0$$

Comparing with $ax^2 + bx + c = 0$, we get

$$a = 2, \quad b = 7, \quad c = -15$$

Using quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-7 \pm \sqrt{7^2 - 4(2)(-15)}}{2(2)}$$

$$x = \frac{-7 \pm \sqrt{49 + 120}}{4}$$

$$x = \frac{-7 \pm \sqrt{169}}{4}$$

$$x = \frac{-7 \pm 13}{4}$$

So,

$$x = \frac{-7 + 13}{4} = \frac{6}{4} = \frac{3}{2}$$

$$x = \frac{-7 - 13}{4} = \frac{-20}{4} = -5$$

THEREFORE,

$$x = \frac{3}{2}, -5$$

(iv) $x^2 + 11 = 7x$

SOLUTION:

Given:

$$x^2 + 11 = 7x$$

Rearranging,

$$x^2 - 7x + 11 = 0$$

Comparing with $ax^2 + bx + c = 0$, we get

$$a = 1, \quad b = -7, \quad c = 11$$

Using quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(-7) \pm \sqrt{(-7)^2 - 4(1)(11)}}{2(1)}$$

$$x = \frac{7 \pm \sqrt{49 - 44}}{2}$$

$$x = \frac{7 \pm \sqrt{5}}{2}$$

So,

$$x = \frac{7 + \sqrt{5}}{2}, \quad x = \frac{7 - \sqrt{5}}{2}$$

Approximately,

$$x \approx 4.618, \quad x \approx 2.382$$

THEREFORE,

$$x = \frac{7 + \sqrt{5}}{2}, \quad \frac{7 - \sqrt{5}}{2}$$

KEY FACTS

COMPLETING SQUARE FORMULA

$$x^2 + bx + c = 0$$

$$\left(x + \frac{b}{2}\right)^2 = \frac{b^2}{4} - c$$

DEFINITION

Completing square method is a method used to solve quadratic equations by making the expression a perfect square.

GENERAL FORM

$$ax^2 + bx + c = 0, \\ a \neq 0$$

CONDITION

$$a \neq 0$$

EXAM TIP

Always take half of the coefficient of x , square it, and add on both sides.

SHORT NOTE

After getting square on one side, take square root on both sides to find the values of x .

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QUESTION 4 Use quadratic formula to solve the following equations:

(vi) $\frac{3x-3}{x+1} = \frac{2x-1}{x-1}$, $x \neq -1, 1$

SOLUTION:

Given: $\frac{3x-3}{x+1} = \frac{2x-1}{x-1}$, $x \neq -1, 1$

STEP 1: Take LCM of denominators

$$\text{LCM of } (x+1) \text{ and } (x-1) = (x+1)(x-1) = x^2 - 1 \quad (\because x \neq -1, 1)$$

Multiply both sides by $(x^2 - 1)$:

$$(3x-3)(x-1) = (2x-1)(x+1)$$

STEP 2: Expand both sides

$$3x(x-1) - 3(x-1) = 2x(x+1) - 1(x+1)$$

$$3x^2 - 3x - 3x + 3 = 2x^2 + 2x - x - 1$$

$$3x^2 - 6x + 3 = 2x^2 + x - 1$$

STEP 3: Bring all terms to one side

$$3x^2 - 6x + 3 - (2x^2 + x - 1) = 0$$

$$3x^2 - 6x + 3 - 2x^2 - x + 1 = 0$$

$$x^2 - 7x + 4 = 0$$

STEP 4: Identify coefficients

Comparing with $ax^2 + bx + c = 0$, we get

$$a = 1, \quad b = -7, \quad c = 4$$

STEP 5: Apply quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Substitute values:

$$x = \frac{-(-7) \pm \sqrt{(-7)^2 - 4(1)(4)}}{2(1)}$$

$$x = \frac{7 \pm \sqrt{49 - 16}}{2}$$

$$x = \frac{7 \pm \sqrt{33}}{2}$$

STEP 6: Find x

$$x = \frac{7 + \sqrt{33}}{2} \quad \text{or} \quad x = \frac{7 - \sqrt{33}}{2}$$

Check: $x \neq -1, 1$ (Both values are valid.)

THEREFORE,

$$x = \frac{7 + \sqrt{33}}{2}, \quad \frac{7 - \sqrt{33}}{2}$$

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QUESTION 5 Solve the following quadratic equations graphically:

(i) $x^2 - 3x - 18 = 0$

SOLUTION:

Given:

$$x^2 - 3x - 18 = 0$$

STEP 1: Write in the form $y = ax^2 + bx + c$

$$y = x^2 - 3x - 18$$

Here, $a = 1$, $b = -3$, $c = -18$

STEP 2: Make table of values

Choose easy and sequence wise values of x .

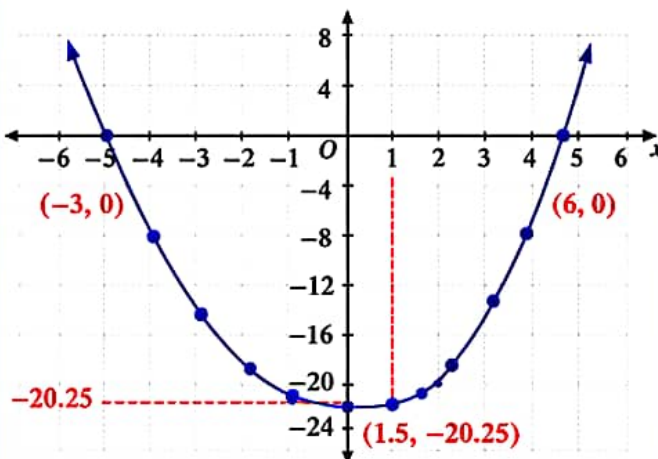
x	-3	-2	-1	0	1	2	3	4	5
y	0	-8	-14	-18	-20	-20	-18	-14	-8

STEP 3: Plot the points

Plot the points $(-3, 0)$, $(-2, -8)$, $(-1, -14)$, $(0, -18)$, $(1, -20)$, $(2, -20)$, $(3, -18)$, $(4, -14)$, $(5, -8)$, $(6, 0)$ on the graph paper.

STEP 4: Draw the graph of $y = x^2 - 3x - 18$

A smooth upward opening parabola is obtained which cuts the x -axis at two points.



STEP 5: Read the solutions

The graph cuts the x -axis at $x = -3$ and $x = 6$.

Check:

For $x = -3$: $(-3)^2 - 3(-3) - 18 = 9 + 9 - 18 = 0$

For $x = 6$: $6^2 - 3(6) - 18 = 36 - 18 - 18 = 0$

THEREFORE,

$$x = -3, 6$$

(ii) $x^2 - 5x - 14 = 0$

SOLUTION:

Given:

$$x^2 - 5x - 14 = 0$$

STEP 1: Write in the form $y = ax^2 + bx + c$

$$y = x^2 - 5x - 14$$

Here, $a = 1$, $b = -5$, $c = -14$

STEP 2: Make table of values

Choose easy and sequence wise values of x .

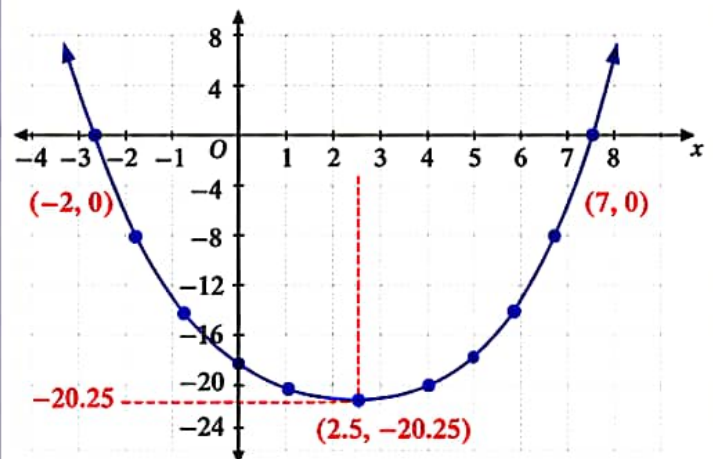
x	-2	-1	0	1	2	3	4	5	6	7
y	0	-8	-14	-18	-20	-20	-18	-14	-8	0

STEP 3: Plot the points

Plot the points $(-2, 0)$, $(-1, -8)$, $(0, -14)$, $(1, -18)$, $(2, -20)$, $(3, -20)$, $(4, -18)$, $(5, -14)$, $(6, -8)$, $(7, 0)$ on the graph paper.

STEP 4: Draw the graph of $y = x^2 - 5x - 14$

A smooth upward opening parabola is obtained which cuts the x -axis at two points.



STEP 5: Read the solutions

The graph cuts the x -axis at $x = -2$ and $x = 7$.

Check:

For $x = -2$: $(-2)^2 - 5(-2) - 14 = 4 + 10 - 14 = 0$

For $x = 7$: $7^2 - 5(7) - 14 = 49 - 35 - 14 = 0$

THEREFORE,

$$x = -2, 7$$

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QUESTION 5 Solve the following quadratic equations graphically:

(iii) $2x^2 + 13x + 6 = 0$

SOLUTION:

Given:

$$2x^2 + 13x + 6 = 0$$

STEP 1: Write in the form $y = ax^2 + bx + c$

$$y = 2x^2 + 13x + 6$$

Here, $a = 2$, $b = 13$, $c = 6$

STEP 2: Make table of values

Choose easy and sequence wise values of x .

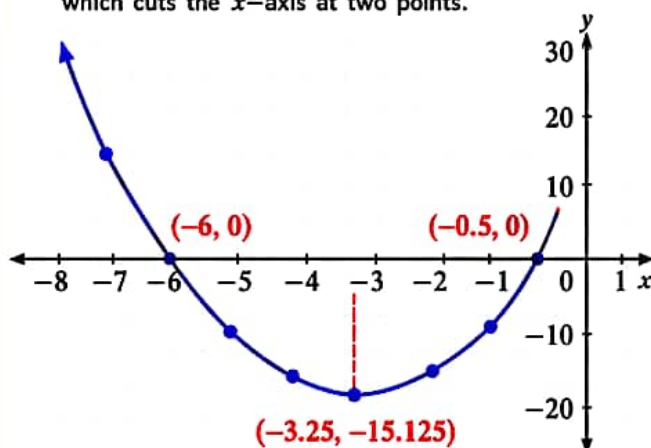
x	-8	-7	-6	-5	-4	-3	-2	-1	-0.5	0
y	30	13	0	-9	-14	-15	-12	-5	0	6

STEP 3: Plot the points

Plot the points $(-8, 30)$, $(-7, 13)$, $(-6, 0)$, $(-5, -9)$, $(-4, -14)$, $(-3, -15)$, $(-2, -12)$, $(-1, -5)$, $(-0.5, 0)$, $(0, 6)$ on the graph paper.

STEP 4: Draw the graph of $y = 2x^2 + 13x + 6$

A smooth upward opening parabola is obtained which cuts the x -axis at two points.



STEP 5: Read the solutions

The graph cuts the x -axis at $x = -6$ and $x = -0.5$.

Check:

For $x = -6$: $2(-6)^2 + 13(-6) + 6 = 72 - 78 + 6 = 0$

For $x = -0.5$: $2(-0.5)^2 + 13(-0.5) + 6 = 0.5 - 6.5 + 6 = 0$

THEREFORE,

$$x = -6, -0.5$$

(iv) $4x^2 + 12x - 27 = 0$

SOLUTION:

Given:

$$4x^2 + 12x - 27 = 0$$

STEP 1: Write in the form $y = ax^2 + bx + c$

$$y = 4x^2 + 12x - 27$$

Here, $a = 4$, $b = 12$, $c = -27$

STEP 2: Make table of values

Choose easy and sequence wise values of x .

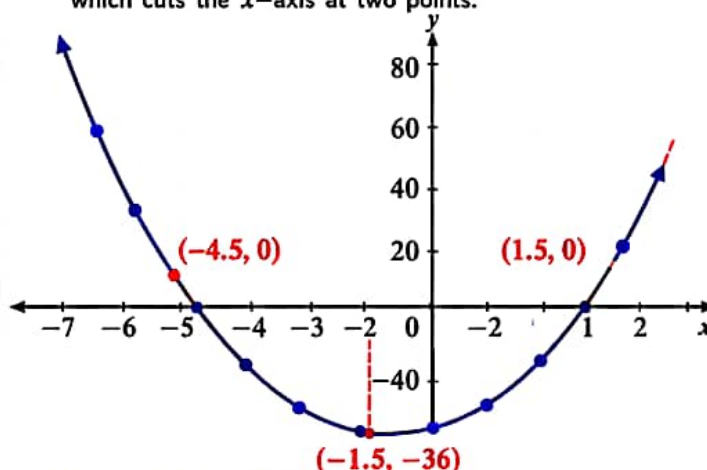
x	-7	-6	-5	-4.5	-4	-3	-2	-1	0	1	1.5	2
y	85	45	13	0	-11	-27	-35	-35	-27	-11	0	13

STEP 3: Plot the points

Plot the points $(-7, 85)$, $(-6, 45)$, $(-5, 13)$, $(-4.5, 0)$, $(-4, -11)$, $(-3, -27)$, $(-2, -35)$, $(-1, -35)$, $(0, -27)$, $(1, -11)$, $(1.5, 0)$, $(2, 13)$ on the graph paper.

STEP 4: Draw the graph of $y = 4x^2 + 12x - 27$

A smooth upward opening parabola is obtained which cuts the x -axis at two points.



STEP 5: Read the solutions

The graph cuts the x -axis at $x = -4.5$ and $x = 1.5$.

Check:

For $x = -4.5$: $4(-4.5)^2 + 12(-4.5) - 27 = 81 - 54 - 27 = 0$

For $x = 1.5$: $4(1.5)^2 + 12(1.5) - 27 = 9 + 18 - 27 = 0$

THEREFORE,

$$x = -4.5, 1.5$$

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QUESTION 1

Find the points of intersection of the following linear equations with coordinate axes graphically:

(i) $x + y = 8$

SOLUTION:

STEP 1: Write the equation in slope-intercept form.

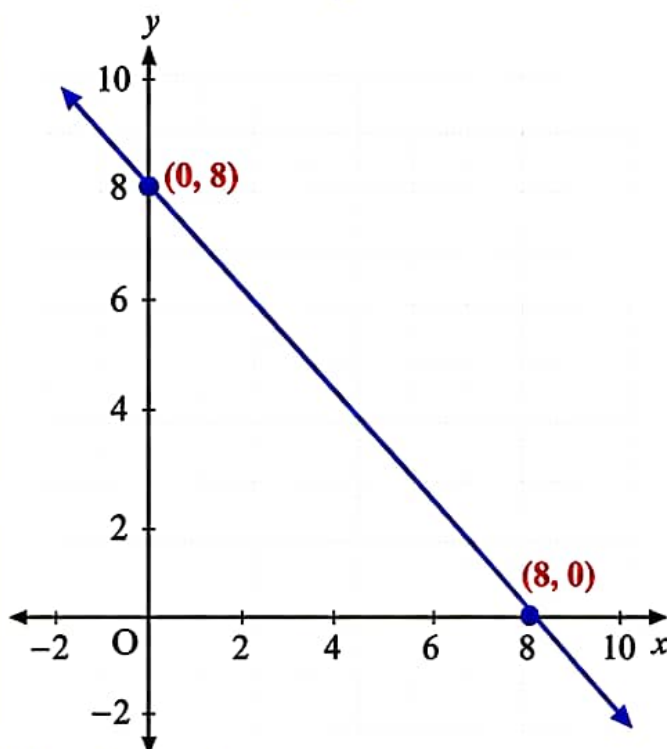
$$x + y = 8 \Rightarrow y = 8 - x$$

STEP 2: Make a table of values.

x	0	8
y	8	0

STEP 3: Plot the points (0, 8) and (8, 0) on the graph.

STEP 4: Join the points to get the line.



From the graph:

$$x - \text{intercept} = (8, 0)$$

$$y - \text{intercept} = (0, 8)$$

THEREFORE, the points of intersection are

(8, 0) and (0, 8)

(ii) $x - y = 1$

SOLUTION:

STEP 1: Write the equation in slope-intercept form.

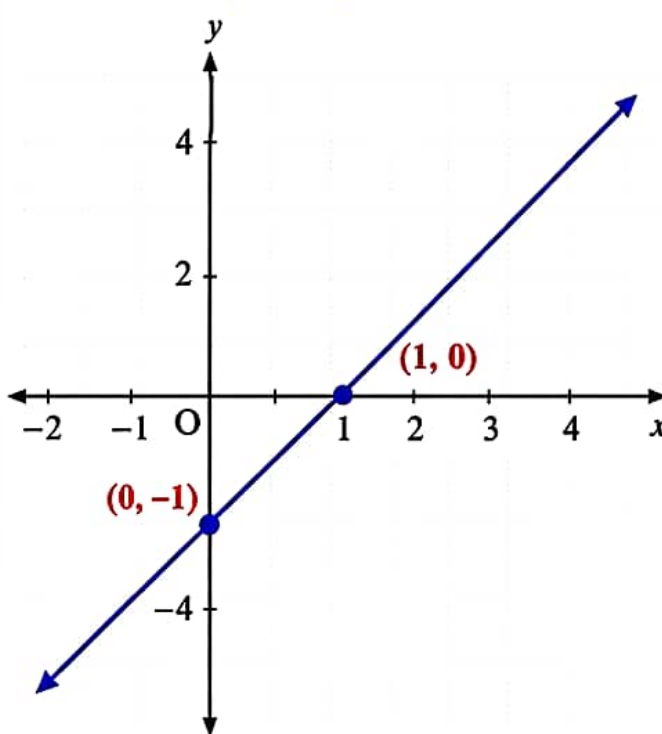
$$x - y = 1 \Rightarrow y = x - 1$$

STEP 2: Make a table of values.

x	0	1
y	-1	0

STEP 3: Plot the points (0, -1) and (1, 0) on the graph.

STEP 4: Join the points to get the line.



From the graph:

$$x - \text{intercept} = (1, 0)$$

$$y - \text{intercept} = (0, -1)$$

THEREFORE, the points of intersection are

(1, 0) and (0, -1)

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QUESTION 1 Find the points of intersection of the following linear equations with coordinate axes graphically:

(iii) $x - 2y = 1$

SOLUTION:

STEP 1: Write the equation in slope-intercept form.

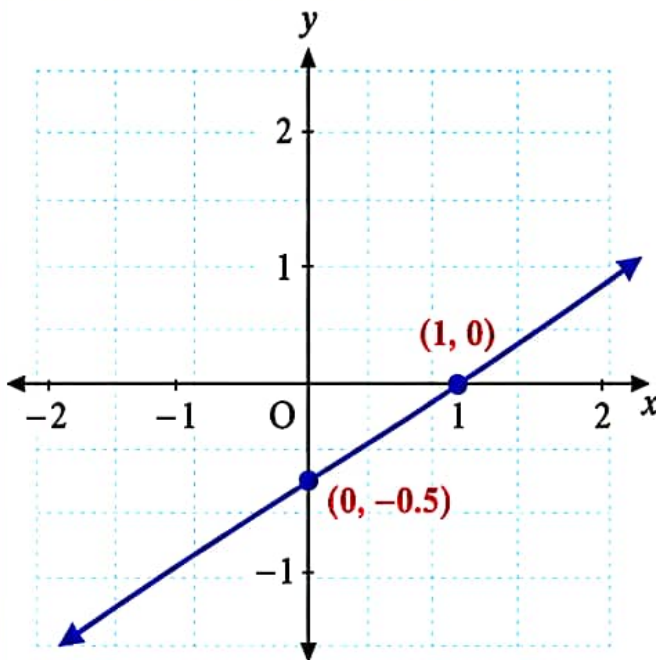
$$x - 2y = 1 \Rightarrow 2y = x - 1 \Rightarrow y = \frac{1}{2}x - \frac{1}{2}$$

STEP 2: Make a table of values.

x	0	1
y	-0.5	0

STEP 3: Plot the points (0, -0.5) and (1, 0) on the graph.

STEP 4: Join the points to get the line.



From the graph:

x - intercept = (1, 0)

y - intercept = (0, -0.5)

THEREFORE, the points of intersection are

(1, 0) and (0, -0.5)

(iv) $x - 2y + 2 = 0$

SOLUTION:

STEP 1: Write the equation in slope-intercept form.

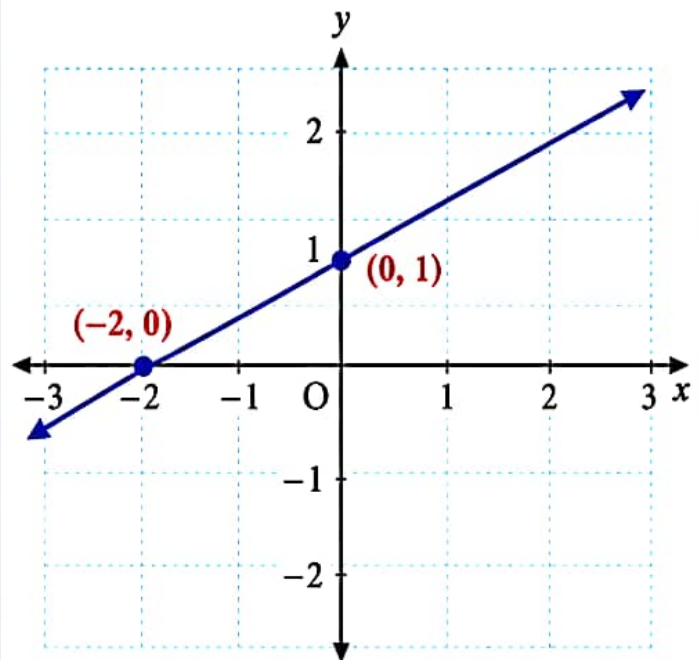
$$x - 2y + 2 = 0 \Rightarrow -2y = -x - 2 \Rightarrow y = \frac{1}{2}x + 1$$

STEP 2: Make a table of values.

x	-2	0
y	0	1

STEP 3: Plot the points (-2, 0) and (0, 1) on the graph.

STEP 4: Join the points to get the line.



From the graph:

x - intercept = (-2, 0)

y - intercept = (0, 1)

THEREFORE, the points of intersection are

(-2, 0) and (0, 1)



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QUESTION 1

Find the points of intersection of the following linear equation with coordinate axes graphically:

(v) $5x - 5y = 1$

SOLUTION:

STEP 1: Write the equation in slope-intercept form.

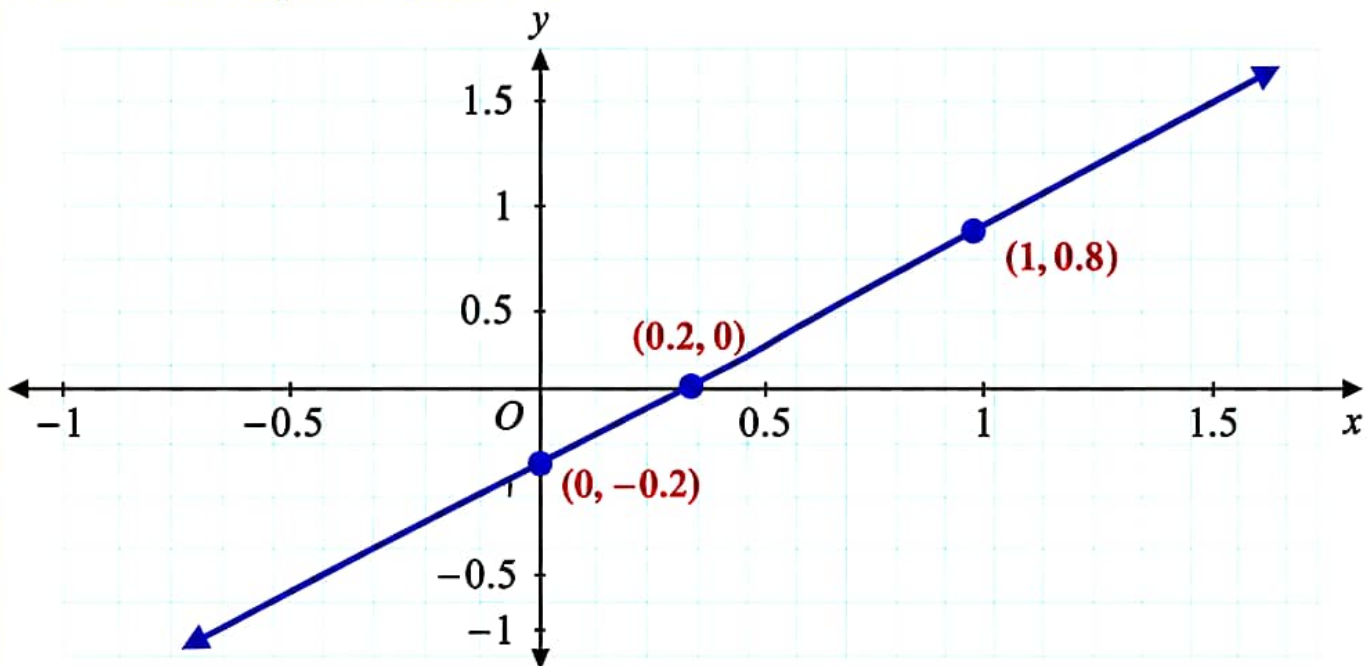
$$5x - 5y = 1 \Rightarrow -5y = 1 - 5x \Rightarrow y = x - \frac{1}{5}$$

STEP 2: Make a table of values.

x	0	0.2	1
y	-0.2	0	0.8

STEP 3: Plot the points $(0, -0.2)$, $(0.2, 0)$ and $(1, 0.8)$ on the graph.

STEP 4: Join the points to get the line.



From the graph:

x - intercept = $(0.2, 0)$

y - intercept = $(0, -0.2)$

THEREFORE, the points of intersection are

$(0.2, 0)$ and $(0, -0.2)$



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QUESTION 2 Solve the following system of linear equations graphically:

(i) $x + y = 8$
 $3x - y = 4$

SOLUTION:

STEP 1: Rewrite the equations in slope-intercept form.

$$x + y = 8 \Rightarrow y = 8 - x \quad \dots(1)$$

$$3x - y = 4 \Rightarrow y = 3x - 4 \quad \dots(2)$$

STEP 2: Make tables of values.

For $y = 8 - x$

x	0	4	8
y	8	4	0

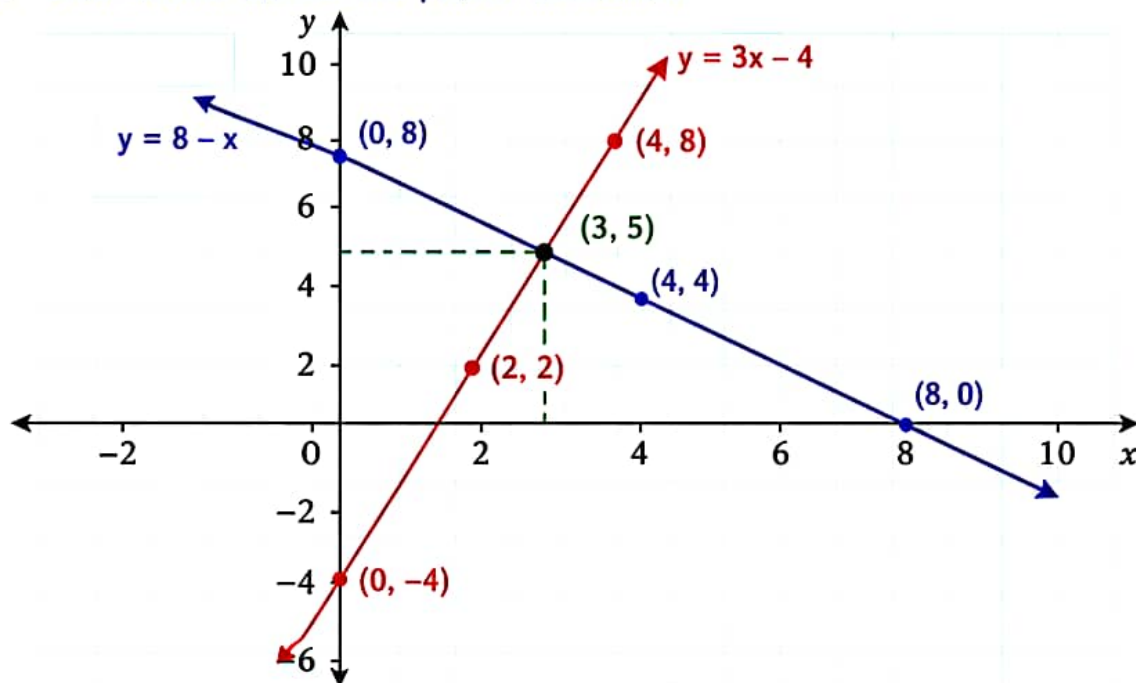
For $y = 3x - 4$

x	0	2	4
y	-4	2	8

STEP 3: Plot the points on the graph.

- For $y = 8 - x$, plot the points (0, 8), (4, 4), (8, 0).
- For $y = 3x - 4$, plot the points (0, -4), (2, 2), (4, 8).

STEP 4: Draw the lines and find their point of intersection.



STEP 5: Read the coordinates of the point of intersection.

The two lines intersect at the point (3, 5).

THEREFORE, the solution of the system of equations is

$$x = 3, y = 5$$



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QUESTION 2 Solve the following system of linear equations graphically:

(ii) $x - y = 1$
 $x + 2y = 7$

SOLUTION:

STEP 1: Rewrite the equations in slope-intercept form.

$$x - y = 1 \Rightarrow y = x - 1 \quad \dots(1)$$

$$x + 2y = 7 \Rightarrow 2y = 7 - x \Rightarrow y = -\frac{1}{2}x + \frac{7}{2} \quad \dots(2)$$

STEP 2: Make tables of values.

For $y = x - 1$

x	0	1	2	3
y	-1	0	1	2

For $y = -\frac{1}{2}x + \frac{7}{2}$

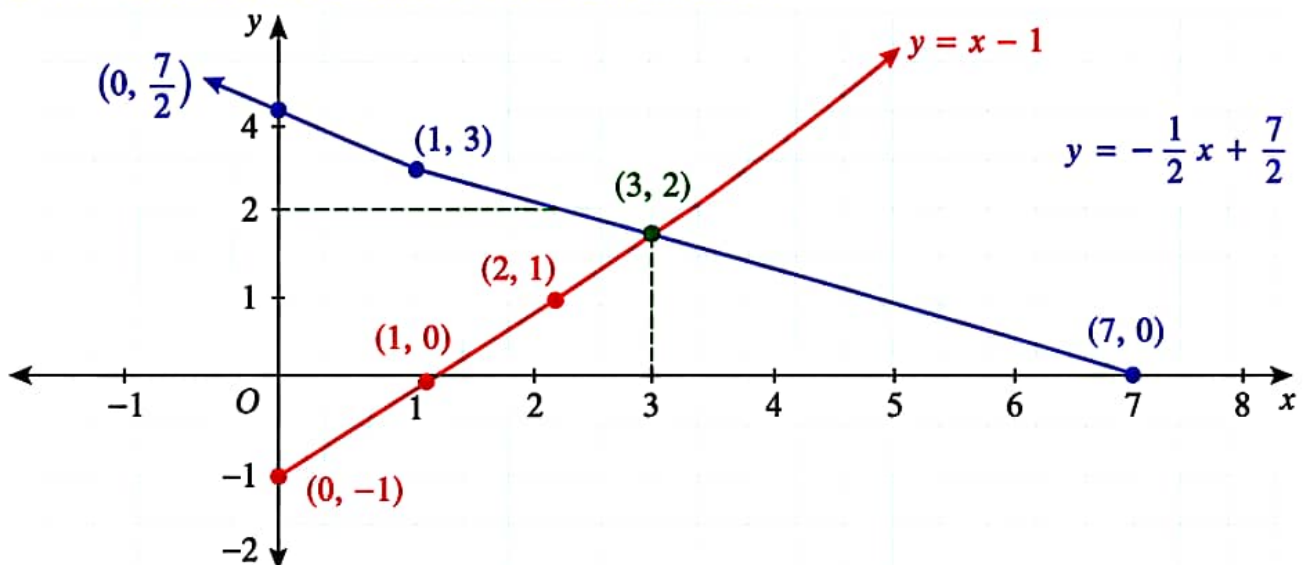
x	0	1	3	7
y	$\frac{7}{2}$	3	2	0

STEP 3: Plot the points on the graph.

- For $y = x - 1$, plot the points $(0, -1)$, $(1, 0)$, $(2, 1)$, $(3, 2)$.

- For $y = -\frac{1}{2}x + \frac{7}{2}$, plot the points $(0, \frac{7}{2})$, $(1, 3)$, $(3, 2)$, $(7, 0)$.

STEP 4: Draw the lines and find their point of intersection.



STEP 5: Read the coordinates of the point of intersection.

The two lines intersect at the point $(3, 2)$.

THEREFORE, the solution of the system of equations is

$$x = 3, y = 2$$



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QUESTION 2 Solve the following system of linear equations graphically:

(iii) $x - 2y = 1$
 $2x + y = 2$

SOLUTION:

STEP 1: Rewrite the equations in slope-intercept form.

$$x - 2y = 1 \Rightarrow -2y = 1 - x \Rightarrow y = \frac{1}{2}x - \frac{1}{2} \quad \dots(1)$$

$$2x + y = 2 \Rightarrow y = 2 - 2x \quad \dots(2)$$

STEP 2: Make tables of values.

For $y = \frac{1}{2}x - \frac{1}{2}$

x	-2	0	2	4
y	$-\frac{3}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$	$\frac{3}{2}$

For $y = 2 - 2x$

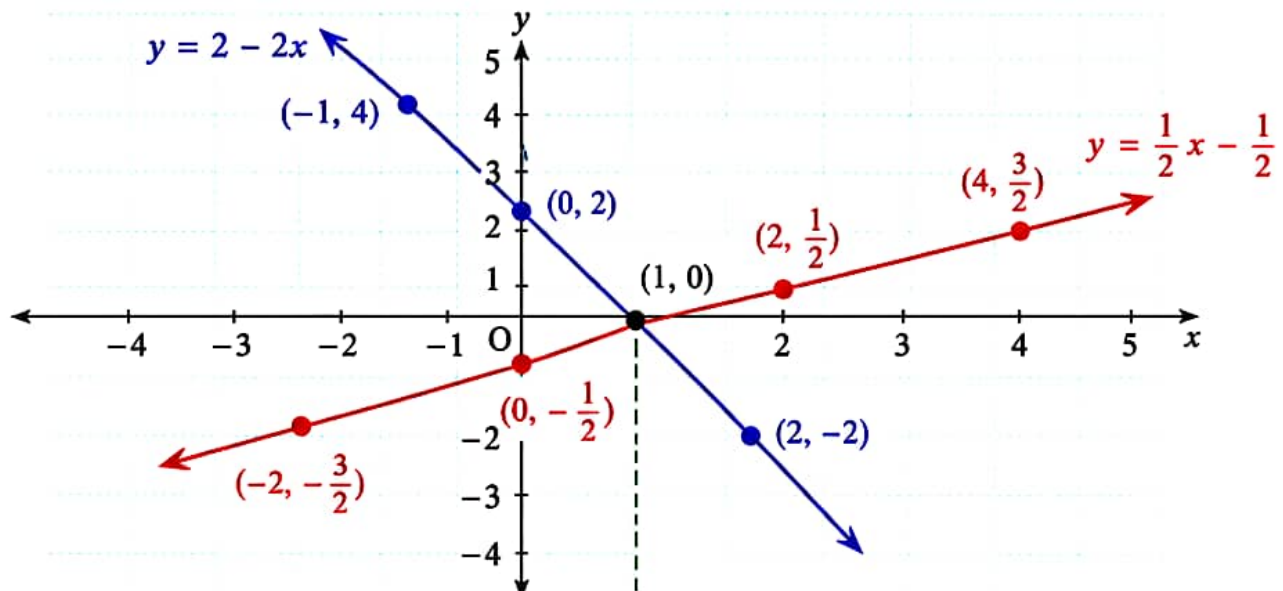
x	-1	0	1	2
y	4	2	0	-2

STEP 3: Plot the points on the graph.

- For $y = \frac{1}{2}x - \frac{1}{2}$, plot the points $(-2, -\frac{3}{2})$, $(0, -\frac{1}{2})$, $(2, \frac{1}{2})$, $(4, \frac{3}{2})$.

- For $y = 2 - 2x$, plot the points $(-1, 4)$, $(0, 2)$, $(1, 0)$, $(2, -2)$.

STEP 4: Draw the lines and find their point of intersection.



STEP 5: Read the coordinates of the point of intersection.

The two lines intersect at the point $(1, 0)$.

THEREFORE, the solution of the system of equations is

$$x = 1, y = 0$$

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QUESTION 2 Solve the following system of linear equations graphically:

(iv) $y = 2x + 2$
 $3x + 2y = 4$

SOLUTION:

STEP 1: Rewrite the equations in slope-intercept form.

$$y = 2x + 2 \quad \dots(1)$$

$$3x + 2y = 4 \Rightarrow 2y = 4 - 3x \Rightarrow y = -\frac{3}{2}x + 2 \quad \dots(2)$$

STEP 2: Make tables of values.

For $y = 2x + 2$

x	-2	-1	0	1	2
y	-2	0	2	4	6

For $y = -\frac{3}{2}x + 2$

x	-2	0	2	4
y	5	2	-1	-4

STEP 3: Plot the points on the graph.

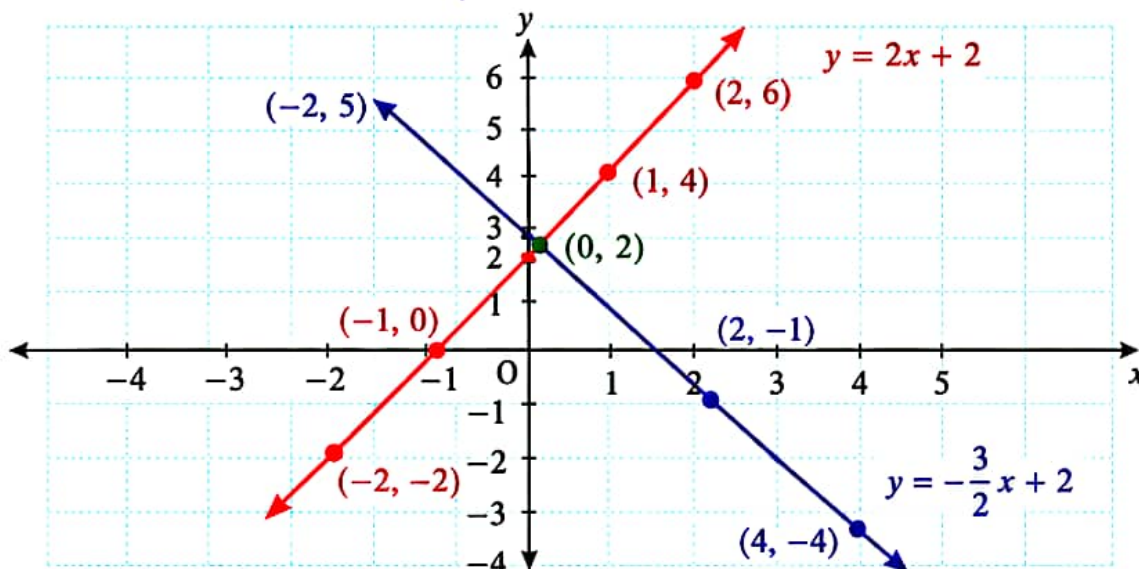
• For $y = 2x + 2$, plot the points

$(-2, -2), (-1, 0), (0, 2), (1, 4), (2, 6)$.

• For $y = -\frac{3}{2}x + 2$, plot the points

$(-2, 5), (0, 2), (2, -1), (4, -4)$.

STEP 4: Draw the lines and find their point of intersection.



STEP 5: Read the coordinates of the point of intersection.

The two lines intersect at the point $(0, 2)$.

THEREFORE, the solution of the system of equations is

$$x = 0, y = 2$$

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QUESTION 2 Solve the following system of linear equations graphically:

$$\begin{aligned} \text{(v)} \quad & 3y = 2x + 8 \\ & x + y = 1 \end{aligned}$$

SOLUTION:

STEP 1: Rewrite the equations in slope-intercept form.

$$3y = 2x + 8 \Rightarrow y = \frac{2}{3}x + \frac{8}{3} \quad \dots(1)$$

$$x + y = 1 \Rightarrow y = 1 - x \quad \dots(2)$$

STEP 2: Make tables of values.

$$\text{For } y = \frac{2}{3}x + \frac{8}{3}$$

x	-4	-1	2	5
y	0	2	4	6

$$\text{For } y = 1 - x$$

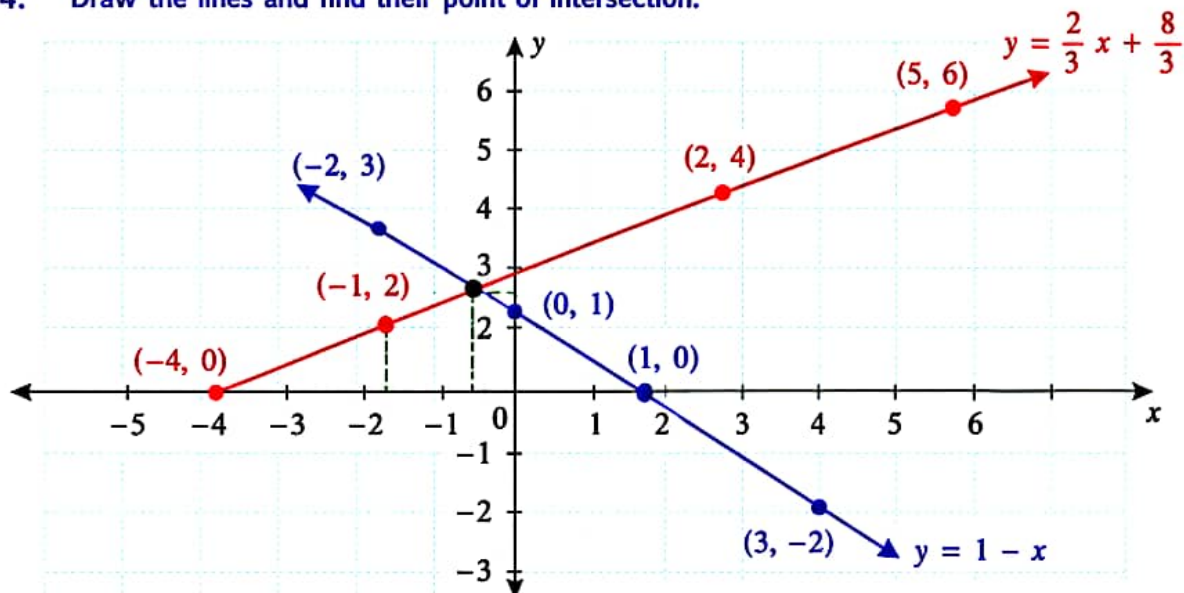
x	-2	0	1	3
y	3	1	0	-2

STEP 3: Plot the points on the graph.

- For $y = \frac{2}{3}x + \frac{8}{3}$, plot the points $(-4, 0)$, $(-1, 2)$, $(2, 4)$, $(5, 6)$.

- For $y = 1 - x$, plot the points $(-2, 3)$, $(0, 1)$, $(1, 0)$, $(3, -2)$.

STEP 4: Draw the lines and find their point of intersection.



STEP 5: Read the coordinates of the point of intersection.

The two lines intersect at the point $(-1, 2)$.

THEREFORE, the solution of the system of equations is $x = -1, y = 2$

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QUESTION 3 (i) Solve the following equations graphically:

$$y = 8x - 32$$

$$y = x^2 - 6x + 8$$

SOLUTION:

STEP 1: Find points to draw the straight line $y = 8x - 32$.

x	0	2	4	6	8	10
y	-32	-16	0	16	32	48

Straight Line Points

Plot the points from the table and draw a straight line.

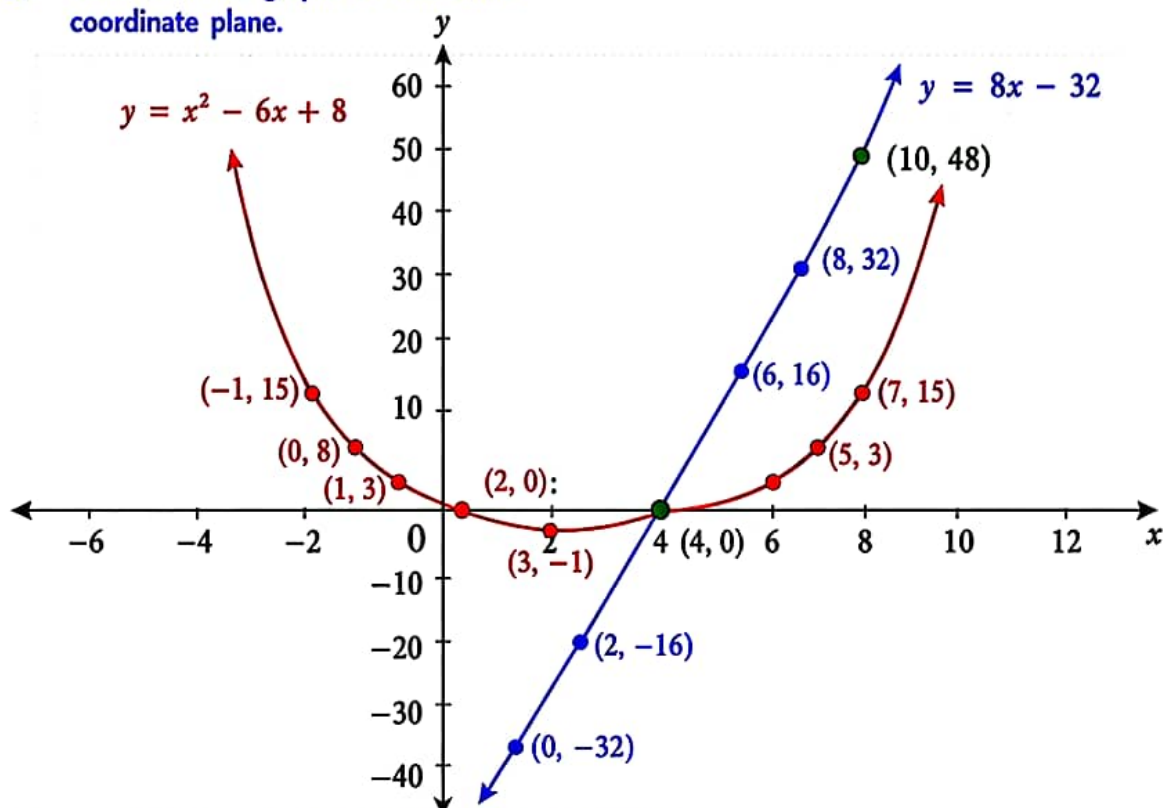
STEP 2: Find points to draw the parabola $y = x^2 - 6x + 8$.

x	-1	0	1	2	3	4	5	6	7
y	15	8	3	0	-1	0	3	8	15

Parabola Points

Plot the points from the table and draw a smooth parabola.

STEP 3: Draw both the graphs on the same coordinate plane.



STEP 4: Read the coordinates of the points of intersection.

The graphs intersect at the points (4, 0) and (10, 48).

THEREFORE, the solutions of the equations are

(4, 0) and (10, 48)

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QUESTION 3 (ii) Solve the following equations graphically:

$$y + x = 2$$

$$y = 2x^2 + x - 10$$

SOLUTION:

STEP 1: Find points to draw the straight line $y + x = 2 \Rightarrow y = 2 - x$.

x	-4	-3	-2	-1	0	1	2	3
y	6	5	4	3	2	1	0	-1

Straight Line Points

Plot the points from the table and draw a straight line.

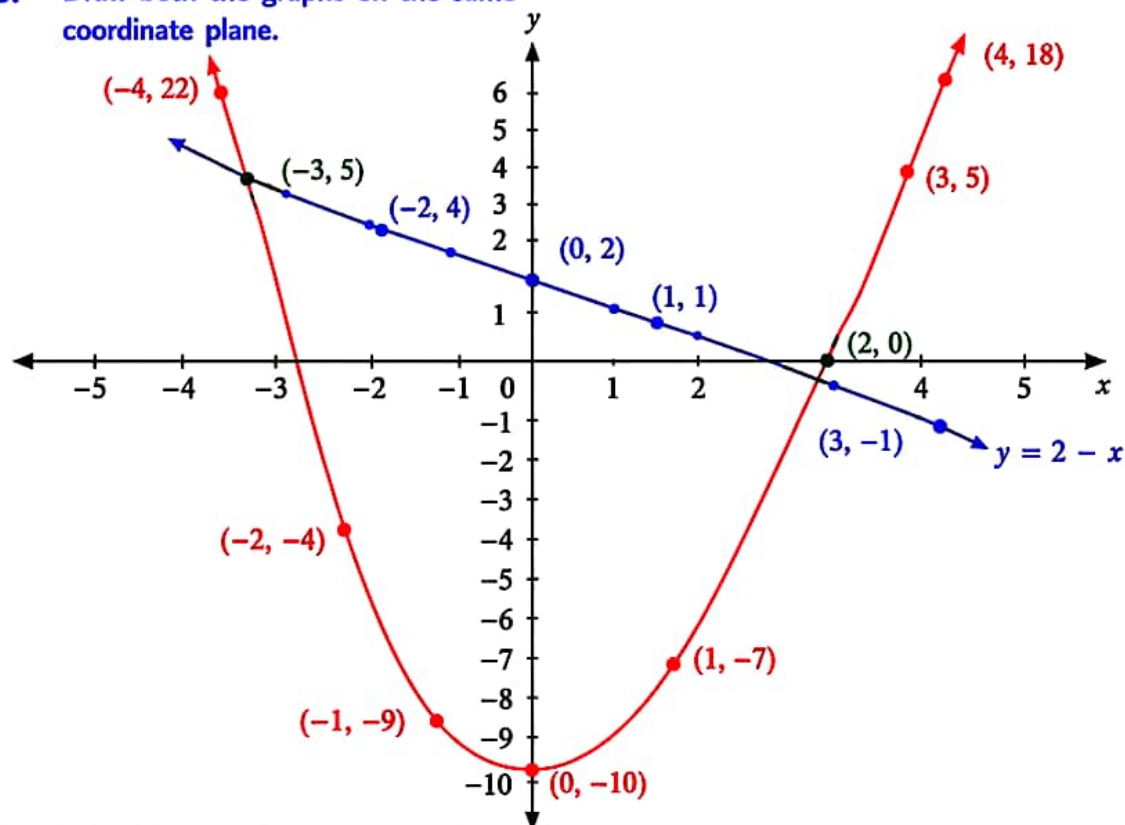
STEP 2: Find points to draw the parabola $y = 2x^2 + x - 10$.

x	-4	-3	-2	-1	0	1	2	3	4
y	22	5	-4	-9	-10	-7	0	5	18

Parabola Points

Plot the points from the table and draw a smooth parabola.

STEP 3: Draw both the graphs on the same coordinate plane.



STEP 4: Read the coordinates of the points of intersection.

The graphs intersect at the points $(-3, 5)$ and $(2, 0)$.

THEREFORE, the solutions of the equations are

$(-3, 5)$ and $(2, 0)$

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QUESTION 1 Form a quadratic equation whose roots are given below:

(i) $-4, 9$

SOLUTION:

Given roots:

$$\alpha = -4, \beta = 9$$

Sum of roots, $S = \alpha + \beta$
 $= -4 + 9 = 5$

Product of roots, $P = \alpha\beta$
 $= (-4)(9) = -36$

Quadratic equation:

$$x^2 - Sx + P = 0$$

$$x^2 - 5x - 36 = 0$$

(ii) $5, -7$

SOLUTION:

Given roots:

$$\alpha = 5, \beta = -7$$

Sum of roots, $S = \alpha + \beta$
 $= 5 + (-7) = -2$

Product of roots, $P = \alpha\beta$
 $= (5)(-7) = -35$

Quadratic equation:

$$x^2 - Sx + P = 0$$

$$x^2 - (-2)x - 35 = 0$$

$$x^2 + 2x - 35 = 0$$

(iii) $\frac{7}{5}, -\frac{6}{5}$

SOLUTION:

Given roots:

$$\alpha = \frac{7}{5}, \beta = -\frac{6}{5}$$

Sum of roots, $S = \alpha + \beta$
 $= \frac{7}{5} + \left(-\frac{6}{5}\right) = \frac{1}{5}$

Product of roots, $P = \alpha\beta$
 $= \left(\frac{7}{5}\right)\left(-\frac{6}{5}\right) = -\frac{42}{25}$

Quadratic equation:

$$x^2 - Sx + P = 0$$

$$x^2 - \left(\frac{1}{5}\right)x - \frac{42}{25} = 0$$

Multiply by 25:

$$25x^2 - 5x - 42 = 0$$

(iv) $-\frac{3}{2}, \frac{7}{2}$

SOLUTION:

Given roots:

$$\alpha = -\frac{3}{2}, \beta = \frac{7}{2}$$

Sum of roots, $S = \alpha + \beta$
 $= -\frac{3}{2} + \frac{7}{2} = \frac{4}{2} = 2$

Product of roots, $P = \alpha\beta$
 $= \left(-\frac{3}{2}\right)\left(\frac{7}{2}\right) = -\frac{21}{4}$

Quadratic equation:

$$x^2 - Sx + P = 0$$

$$x^2 - 2x - \frac{21}{4} = 0$$

Multiply by 4:

$$4x^2 - 8x - 21 = 0$$

(v) $3 + \sqrt{5}, 3 - \sqrt{5}$

SOLUTION:

Given roots:

$$\alpha = 3 + \sqrt{5}, \beta = 3 - \sqrt{5}$$

Sum of roots, $S = \alpha + \beta$
 $= (3 + \sqrt{5}) + (3 - \sqrt{5})$
 $= 6$

Product of roots, $P = \alpha\beta$
 $= (3 + \sqrt{5})(3 - \sqrt{5})$
 $= 3^2 - (\sqrt{5})^2 = 9 - 5 = 4$

Quadratic equation:

$$x^2 - Sx + P = 0$$

$$x^2 - 6x + 4 = 0$$

(vi) $-2 + \sqrt{3}, -2 - \sqrt{3}$

SOLUTION:

Given roots:

$$\alpha = -2 + \sqrt{3}, \beta = -2 - \sqrt{3}$$

Sum of roots, $S = \alpha + \beta$
 $= (-2 + \sqrt{3}) + (-2 - \sqrt{3})$
 $= -4$

Product of roots, $P = \alpha\beta$
 $= (-2 + \sqrt{3})(-2 - \sqrt{3})$
 $= (-2)^2 - (\sqrt{3})^2 = 4 - 3 = 1$

Quadratic equation:

$$x^2 - Sx + P = 0$$

$$x^2 - (-4)x + 1 = 0$$

$$x^2 + 4x + 1 = 0$$

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QUESTION 2

Find the quadratic equation with roots exceeding by 2 than those of roots of $x^2 + 9x + 20 = 0$.

SOLUTION:

Given equation: $x^2 + 9x + 20 = 0$

Let its roots be α and β .

$$\text{Then, } \alpha + \beta = -\frac{b}{a} = -\frac{9}{1} = -9 \quad \text{and} \quad \alpha\beta = \frac{c}{a} = \frac{20}{1} = 20$$

Required roots = $\alpha + 2$ and $\beta + 2$

Sum of new roots, $S' = (\alpha + 2) + (\beta + 2) = (\alpha + \beta) + 4 = -9 + 4 = -5$

Product of new roots, $P' = (\alpha + 2)(\beta + 2) = \alpha\beta + 2(\alpha + \beta) + 4$
 $= 20 + 2(-9) + 4 = 20 - 18 + 4 = 6$

Quadratic equation with roots having sum -5 and product 6 is

$$\begin{aligned} x^2 - S'x + P' &= 0 \\ \Rightarrow x^2 - (-5)x + 6 &= 0 \end{aligned}$$

$$x^2 + 5x + 6 = 0$$

QUESTION 3

Find the equation whose roots are double the roots of $x^2 - px + q = 0$.

SOLUTION:

Given equation: $x^2 - px + q = 0$

Let its roots be α and β .

$$\text{Then, } \alpha + \beta = -\frac{b}{a} = \frac{p}{1} = p \quad \text{and} \quad \alpha\beta = \frac{c}{a} = \frac{q}{1} = q$$

Required roots = 2α and 2β

Sum of new roots, $S' = 2\alpha + 2\beta = 2(\alpha + \beta) = 2p$

Product of new roots, $P' = (2\alpha)(2\beta) = 4\alpha\beta = 4q$

Quadratic equation with roots having sum $2p$ and product $4q$ is

$$\begin{aligned} x^2 - S'x + P' &= 0 \\ \Rightarrow x^2 - (2p)x + 4q &= 0 \end{aligned}$$

$$x^2 - 2px + 4q = 0$$



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QUESTION 4 If α, β are the roots of the equation $x^2 + 2x + 4 = 0$, then find the equation whose roots are:

(i) $\frac{1}{\alpha}, \frac{1}{\beta}$

(ii) $\frac{\alpha}{\beta}, \frac{\beta}{\alpha}$

SOLUTION

(i) **Given:** $x^2 + 2x + 4 = 0$

For $x^2 + 2x + 4 = 0 \Rightarrow \alpha + \beta = -2, \alpha\beta = 4$

Required roots: $\frac{1}{\alpha}, \frac{1}{\beta}$

• Sum of new roots, S

$$\begin{aligned} S &= \frac{1}{\alpha} + \frac{1}{\beta} \\ &= \frac{\alpha + \beta}{\alpha\beta} \\ &= \frac{(-2)}{4} = -\frac{1}{2} \end{aligned}$$

• Product of new roots, P

$$\begin{aligned} P &= \frac{1}{\alpha} \cdot \frac{1}{\beta} \\ &= \frac{1}{\alpha\beta} \\ &= \frac{1}{4} \end{aligned}$$

Quadratic equation:

$$\begin{aligned} x^2 - Sx + P &= 0 \\ x^2 - \left(-\frac{1}{2}\right)x + \frac{1}{4} &= 0 \\ x^2 + \frac{1}{2}x + \frac{1}{4} &= 0 \end{aligned}$$

Multiply by 4:

$$4x^2 + 2x + 1 = 0$$

(ii) **Given:** $x^2 + 2x + 4 = 0$

For $x^2 + 2x + 4 = 0 \Rightarrow \alpha + \beta = -2, \alpha\beta = 4$

Required roots: $\frac{\alpha}{\beta}, \frac{\beta}{\alpha}$

• Sum of new roots, S

$$\begin{aligned} S &= \frac{\alpha}{\beta} + \frac{\beta}{\alpha} \\ &= \frac{\alpha^2 + \beta^2}{\alpha\beta} \\ &= \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta} \\ &= \frac{(-2)^2 - 2(4)}{4} \\ &= \frac{4 - 8}{4} = -1 \end{aligned}$$

• Product of new roots, P

$$\begin{aligned} P &= \frac{\alpha}{\beta} \cdot \frac{\beta}{\alpha} \\ &= 1 \end{aligned}$$

Quadratic equation:

$$\begin{aligned} x^2 - Sx + P &= 0 \\ x^2 - (-1)x + 1 &= 0 \\ x^2 + x + 1 &= 0 \end{aligned}$$

$$x^2 + x + 1 = 0$$

KEY POINT

If α, β are the roots of $ax^2 + bx + c = 0$, then

$$\alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}$$

Note:

If new roots are expressed in terms of α and β , find their sum (S) and product (P), then use $x^2 - Sx + P = 0$ to form the required equation.

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QUESTION 4 If α, β are the roots of the equation $x^2 + 2x + 4 = 0$, then find the equation whose roots are:

(iii) $2\alpha - \frac{1}{\beta}, 2\beta - \frac{1}{\alpha}$

(iv) α^2, β^2

(v) $2\alpha - 1, 2\beta - 1$

SOLUTION

(iii) **Given:** $x^2 + 2x + 4 = 0$

For $x^2 + 2x + 4 = 0 \Rightarrow \alpha + \beta = -2, \alpha\beta = 4$

Required roots: $2\alpha - \frac{1}{\beta}, 2\beta - \frac{1}{\alpha}$

• **Sum of new roots, S**

$$\begin{aligned} S &= \left(2\alpha - \frac{1}{\beta}\right) + \left(2\beta - \frac{1}{\alpha}\right) \\ &= 2(\alpha + \beta) - \left(\frac{1}{\beta} + \frac{1}{\alpha}\right) \\ &= 2(-2) - \frac{\alpha + \beta}{\alpha\beta} \\ &= -4 - \frac{-2}{4} = -4 + \frac{1}{2} = -\frac{7}{2} \end{aligned}$$

• **Product of new roots, P**

$$\begin{aligned} P &= \left(2\alpha - \frac{1}{\beta}\right) \left(2\beta - \frac{1}{\alpha}\right) \\ &= 4\alpha\beta - 2\left(\frac{\alpha}{\beta} + \frac{\beta}{\alpha}\right) + \frac{1}{\alpha\beta} \\ &= 4(4) - 2\left(1 + 1\right) + \frac{1}{4} \\ &= 16 - 4 + \frac{1}{4} = \frac{49}{4} \end{aligned}$$

Quadratic equation:

$$\begin{aligned} x^2 - Sx + P &= 0 \\ x^2 - \left(-\frac{7}{2}\right)x + \frac{49}{4} &= 0 \\ x^2 + \frac{7}{2}x + \frac{49}{4} &= 0 \end{aligned}$$

Multiply by 4:

$$4x^2 + 14x + 49 = 0$$

(iv) **Given:** $x^2 + 2x + 4 = 0$

$\Rightarrow \alpha + \beta = -2, \alpha\beta = 4$

Required roots: α^2, β^2

• **Sum of new roots, S**

$$\begin{aligned} S &= \alpha^2 + \beta^2 \\ &= (\alpha + \beta)^2 - 2\alpha\beta \\ &= (-2)^2 - 2(4) \\ &= 4 - 8 = -4 \end{aligned}$$

• **Product of new roots, P**

$$\begin{aligned} P &= \alpha^2\beta^2 = (\alpha\beta)^2 \\ &= (4)^2 = 16 \end{aligned}$$

Quadratic equation:

$$\begin{aligned} x^2 - Sx + P &= 0 \\ x^2 - (-4)x + 16 &= 0 \\ x^2 + 4x + 16 &= 0 \end{aligned}$$

$$x^2 + 4x + 16 = 0$$

(v) **Given:** $x^2 + 2x + 4 = 0$

$\Rightarrow \alpha + \beta = -2, \alpha\beta = 4$

Required roots: $2\alpha - 1, 2\beta - 1$

• **Sum of new roots, S**

$$\begin{aligned} S &= (2\alpha - 1) + (2\beta - 1) \\ &= 2(\alpha + \beta) - 2 \\ &= 2(-2) - 2 = -4 - 2 = -6 \end{aligned}$$

• **Product of new roots, P**

$$\begin{aligned} P &= (2\alpha - 1)(2\beta - 1) \\ &= 4\alpha\beta - 2(\alpha + \beta) + 1 \\ &= 4(4) - 2(-2) + 1 \\ &= 16 + 4 + 1 = 21 \end{aligned}$$

Quadratic equation:

$$\begin{aligned} x^2 - Sx + P &= 0 \\ x^2 - (-6)x + 21 &= 0 \\ x^2 + 6x + 21 &= 0 \end{aligned}$$

$$x^2 + 6x + 21 = 0$$

QUESTION 5 Find the condition that roots of $ax^2 + bx + c = 0$ should be reciprocals of each other.

SOLUTION:

Let the roots be α and β .

If roots are reciprocals of each other, then

$$\alpha = \frac{1}{\beta} \Rightarrow \alpha\beta = 1$$

For $ax^2 + bx + c = 0$, $\alpha\beta = \frac{c}{a}$

Since $\alpha\beta = 1 \Rightarrow \frac{c}{a} = 1 \Rightarrow c = a \Rightarrow \mathbf{a = c}$

KEY POINT

If the roots of a quadratic equation are reciprocals of each other, their product is 1. Therefore, $c = a$.

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QUESTION 6 Find the value of k , given that one root of $x^2 - (2k + 4)x + (7k + 1) = 0$ is 3.

SOLUTION:

Let $x^2 - (2k + 4)x + (7k + 1) = 0$ and one root is 3.

Since 3 is a root, substitute $x = 3$ in the equation.

$$3^2 - (2k + 4)(3) + (7k + 1) = 0$$

$$\Rightarrow 9 - 3(2k + 4) + 7k + 1 = 0$$

$$\Rightarrow 9 - (6k + 12) + 7k + 1 = 0$$

$$\Rightarrow 9 - 6k - 12 + 7k + 1 = 0$$

$$\Rightarrow (-6k + 7k) + (9 - 12 + 1) = 0$$

$$\Rightarrow k - 2 = 0$$

$$\Rightarrow \boxed{k = 2}$$

KEY POINT



If α is a root of

$$ax^2 + bx + c = 0,$$

then $a\alpha^2 + b\alpha + c = 0$.

QUESTION 7 Find the value of m in the equation $2x^2 + 3x + m = 0$ when sum of its roots is equal to double the product of its roots.

SOLUTION:

Let $2x^2 + 3x + m = 0$ have roots α and β .

$$\text{Sum of the roots, } S = \alpha + \beta = -\frac{b}{a} = -\frac{3}{2} \quad [\because a = 2, b = 3]$$

$$\text{Product of the roots, } P = \alpha\beta = \frac{c}{a} = \frac{m}{2} \quad [\because c = m]$$

According to the question,

$$\boxed{S = 2P}$$

$$-\frac{3}{2} = 2 \left(\frac{m}{2} \right)$$

$$-\frac{3}{2} = m$$

$$\boxed{m = -\frac{3}{2}}$$

FORMULAS USED

For $ax^2 + bx + c = 0$

$$\boxed{S = \alpha + \beta = -\frac{b}{a}}$$

$$\boxed{P = \alpha\beta = \frac{c}{a}}$$

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QUESTION 8 If α, β are the roots of $x^2 + ax + b = 0$ and α^2, β^2 are the roots of $x^2 + Ax + B = 0$, then prove that $A = 2b - a^2$, $B = b^2$.

ANSWER:

For $x^2 + ax + b = 0$

Since α, β are the roots,

$$\alpha + \beta = -a, \quad \alpha\beta = b \quad \dots(1)$$

For $x^2 + Ax + B = 0$

Since α^2, β^2 are the roots,

$$\alpha^2 + \beta^2 = -A, \quad \alpha^2\beta^2 = B \quad \dots(2)$$

From (1),

$$\begin{aligned}(\alpha + \beta)^2 &= \alpha^2 + \beta^2 + 2\alpha\beta \\ \Rightarrow \alpha^2 + \beta^2 &= (\alpha + \beta)^2 - 2\alpha\beta \\ &= (-a)^2 - 2(b) \\ &= a^2 - 2b\end{aligned}$$

From (2), $\alpha^2 + \beta^2 = -A$

$$\Rightarrow -A = a^2 - 2b$$

$$A = 2b - a^2$$

Also, $\alpha^2\beta^2 = (\alpha\beta)^2$

$$= (b)^2 \quad [\text{from (1)}]$$

$$= b^2$$

From (2), $\alpha^2\beta^2 = B$

$$\Rightarrow B = b^2$$

Hence proved: $A = 2b - a^2$, $B = b^2$

QUESTION 9 If α, β are the roots of $x^2 + px + q = 0$, then find the condition that

(i) $\alpha = \beta$

(ii) $\alpha = \frac{1}{\beta}$

ANSWER:

Since α, β are the roots,

$$\alpha + \beta = -p, \quad \alpha\beta = q \quad \dots(1)$$

(i) Condition for $\alpha = \beta$

If $\alpha = \beta$, then $\alpha + \beta = \alpha + \alpha = 2\alpha$

$$\alpha\beta = \alpha \cdot \alpha = \alpha^2$$

$$\text{From (1), } -p = 2\alpha \Rightarrow \alpha = -\frac{p}{2}$$

$$\text{From (1), } q = \alpha^2 = \left(-\frac{p}{2}\right)^2 = \frac{p^2}{4}$$

$$\text{Condition: } q = \frac{p^2}{4}$$

(ii) Condition for $\alpha = \frac{1}{\beta}$

If $\alpha = \frac{1}{\beta}$, then

$$\alpha\beta = \frac{1}{\beta}, \quad \beta = 1$$

$$\text{From (1), } q = 1$$

$$\text{Condition: } q = 1$$

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EXERCISE 2.4

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QUESTION 1

Examine the nature of roots of the following quadratic equations:

(i) $3x^2 - 9x - 2 = 0$ (ii) $x^2 + 6x + 9 = 0$ (iii) $2x^2 + 4x + 5 = 0$

SOLUTION:



We use discriminant $D = b^2 - 4ac$ to examine the nature of roots.

- If $D > 0$ but not a perfect square $\Rightarrow$ roots are irrational and unequal.
- If $D = 0$ $\Rightarrow$ roots are real and equal.
- If $D > 0$ and a perfect square $\Rightarrow$ roots are rational and unequal.
- If $D < 0$ $\Rightarrow$ roots are imaginary.

(i) $3x^2 - 9x - 2 = 0$

Here, $a = 3$, $b = -9$, $c = -2$

$$\begin{aligned}\text{Disc.} &= b^2 - 4ac \\ &= (-9)^2 - 4(3)(-2) \\ &= 81 + 24 \\ &= 105 > 0\end{aligned}$$

Since 105 is not a perfect square.

Conclusion

Discriminant is positive
but not a perfect square.
Therefore, the roots are

irrational and unequal.

(ii) $x^2 + 6x + 9 = 0$

Here, $a = 1$, $b = 6$, $c = 9$

$$\begin{aligned}\text{Disc.} &= b^2 - 4ac \\ &= (6)^2 - 4(1)(9) \\ &= 36 - 36 \\ &= 0\end{aligned}$$

Since discriminant is zero.

Conclusion

Discriminant is zero.
Therefore, the roots are

real and equal.

(iii) $2x^2 + 4x + 5 = 0$

Here, $a = 2$, $b = 4$, $c = 5$

$$\begin{aligned}\text{Disc.} &= b^2 - 4ac \\ &= (4)^2 - 4(2)(5) \\ &= 16 - 40 \\ &= -24 < 0\end{aligned}$$

Since discriminant is negative.

Conclusion

Discriminant is negative.
Therefore, the roots are

imaginary.

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QUESTION 1

Examine the nature of roots of the following quadratic equations:

(iv) $7x^2 - 6x - 1 = 0$ (v) $5x^2 - 2x + 10 = 0$ (vi) $x^2 - 8x + 16 = 0$

SOLUTION:



We use discriminant $D = b^2 - 4ac$ to examine the nature of roots.

- If $D > 0$ but not a perfect square $\Rightarrow$ roots are irrational and unequal.
- If $D = 0$ $\Rightarrow$ roots are real and equal.
- If $D > 0$ and a perfect square $\Rightarrow$ roots are rational and unequal.
- If $D < 0$ $\Rightarrow$ roots are imaginary.

(iv) $7x^2 - 6x - 1 = 0$

Here, $a = 7$, $b = -6$, $c = -1$

$$\begin{aligned}\text{Disc.} &= b^2 - 4ac \\ &= (-6)^2 - 4(7)(-1) \\ &= 36 + 28 \\ &= 64 > 0\end{aligned}$$

Since 64 is a perfect square.

Conclusion

Discriminant is positive
and a perfect square.
Therefore, the roots are

rational and unequal.

(v) $5x^2 - 2x + 10 = 0$

Here, $a = 5$, $b = -2$, $c = 10$

$$\begin{aligned}\text{Disc.} &= b^2 - 4ac \\ &= (-2)^2 - 4(5)(10) \\ &= 4 - 200 \\ &= -196 < 0\end{aligned}$$

Since discriminant is negative.

Conclusion

Discriminant is negative.
Therefore, the roots are

imaginary.

(vi) $x^2 - 8x + 16 = 0$

Here, $a = 1$, $b = -8$, $c = 16$

$$\begin{aligned}\text{Disc.} &= b^2 - 4ac \\ &= (-8)^2 - 4(1)(16) \\ &= 64 - 64 \\ &= 0\end{aligned}$$

Since discriminant is zero.

Conclusion

Discriminant is zero.
Therefore, the roots are

real and equal.



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QUESTION 2 For what values of t , the roots of $3x^2 + x + 9t = 0$ are real and unequal?

ANSWER:

Given equation: $3x^2 + x + 9t = 0$

Here, $a = 3$, $b = 1$, $c = 9t$

For **real and unequal** roots:

$$D = b^2 - 4ac > 0$$

Now, $D = (1)^2 - 4(3)(9t) > 0$

$$1 - 108t > 0$$

$$-108t > -1$$

$$t < \frac{1}{108}$$

Conclusion

The roots of $3x^2 + x + 9t = 0$
are real and unequal if

$$t < \frac{1}{108}$$

QUESTION 3 If the quadratic equation $16x^2 + 7px + 49 = 0$ has equal roots, then find the values of p .

ANSWER:

Given equation: $16x^2 + 7px + 49 = 0$

Here, $a = 16$, $b = 7p$, $c = 49$

For **equal** roots:

$$D = b^2 - 4ac = 0$$

Now, $(7p)^2 - 4(16)(49) = 0$

$$49p^2 - 3136 = 0$$

$$49p^2 = 3136$$

$$p^2 = \frac{3136}{49}$$

$$p^2 = 64$$

$$p = \pm 8$$

Conclusion

The quadratic equation
 $16x^2 + 7px + 49 = 0$
has equal roots for

$$p = \pm 8$$

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QUESTION 4

If the quadratic equation $4u^2 + 8u + q = 0$ has unequal and real roots, find the possible values for q .

SOLUTION:

Given equation: $4u^2 + 8u + q = 0$

Here, $a = 4$, $b = 8$, $c = q$

For unequal and real roots,

$$D = b^2 - 4ac > 0$$

$$(8)^2 - 4(4)(q) > 0$$

$$64 - 16q > 0$$

$$-16q > -64$$

$$q < \frac{-64}{-16}$$

$$q < 4$$

Conclusion

For unequal and real roots,
the possible values of q are

$$q < 4$$

QUESTION 5

Find the value for m , if the quadratic equation $mx^2 - 8x + 1 = 0$ has real and equal roots.

SOLUTION:

Given equation: $mx^2 - 8x + 1 = 0$

Here, $a = m$, $b = -8$, $c = 1$

For real and equal roots,

$$D = b^2 - 4ac = 0$$

$$(-8)^2 - 4(m)(1) = 0$$

$$64 - 4m = 0$$

$$64 = 4m$$

$$m = \frac{64}{4}$$

$$m = 16$$

Conclusion

The value of m for real
and equal roots is

$$m = 16$$

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EXERCISE 2.5

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QUESTION 1 Solve the following inequalities:

(i) $x^2 + 3x - 4 > 0$

(ii) $2x^2 - 8x + 6 > 0$

SOLUTION:

(i) $x^2 + 3x - 4 > 0$

Step I Solve the corresponding equation.

$$x^2 + 3x - 4 = 0$$

$$(x + 4)(x - 1) = 0$$

$$x + 4 = 0 \quad \text{or} \quad x - 1 = 0$$

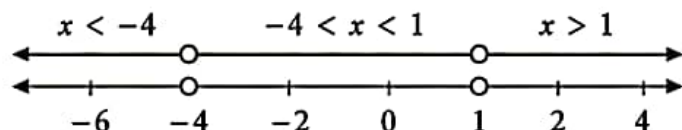
$$x = -4, 1$$

Step II Determine intervals.

Interval 1: $x < -4$

Interval 2: $-4 < x < 1$

Interval 3: $x > 1$



Step III Test points in each interval.

For $x = -5$ (in $x < -4$):

$$(-5)^2 + 3(-5) - 4 = 25 - 15 - 4 = 6 > 0 \quad \checkmark$$

For $x = 0$ (in $-4 < x < 1$):

$$(0)^2 + 3(0) - 4 = -4 < 0 \quad \times$$

For $x = 2$ (in $x > 1$):

$$(2)^2 + 3(2) - 4 = 4 + 6 - 4 = 6 > 0 \quad \checkmark$$

Step IV Conclusion:

The solution of the inequality is

$$(-\infty, -4) \cup (1, \infty)$$

(ii) $2x^2 - 8x + 6 > 0$

Step I Solve the corresponding equation.

$$2x^2 - 8x + 6 = 0$$

$$2(x^2 - 4x + 3) = 0$$

$$2(x - 1)(x - 3) = 0$$

$$x - 1 = 0 \quad \text{or} \quad x - 3 = 0$$

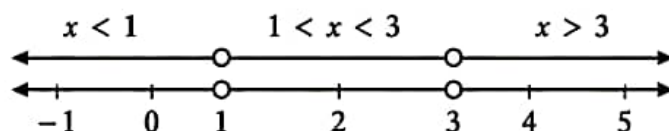
$$x = 1, 3$$

Step II Determine intervals.

Interval 1: $x < 1$

Interval 2: $1 < x < 3$

Interval 3: $x > 3$



Step III Test points in each interval.

For $x = 0$ (in $x < 1$):

$$2(0)^2 - 8(0) + 6 = 6 > 0 \quad \checkmark$$

For $x = 2$ (in $1 < x < 3$):

$$2(2)^2 - 8(2) + 6 = 8 - 16 + 6 = -2 < 0 \quad \times$$

For $x = 4$ (in $x > 3$):

$$2(4)^2 - 8(4) + 6 = 32 - 32 + 6 = 6 > 0 \quad \checkmark$$

Step IV Conclusion:

The solution of the inequality is

$$(-\infty, 1) \cup (3, \infty)$$

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EXERCISE 2.5

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QUESTION 1

Solve the following inequalities:

(iii) $x^2 + x - 6 < 0$

(iv) $x^2 - 6x + 9 < 0$

SOLUTION:

(iii) $x^2 + x - 6 < 0$

Step I Solve the corresponding equation.

$$x^2 + x - 6 = 0$$

$$(x + 3)(x - 2) = 0$$

$$x + 3 = 0 \quad \text{or} \quad x - 2 = 0$$

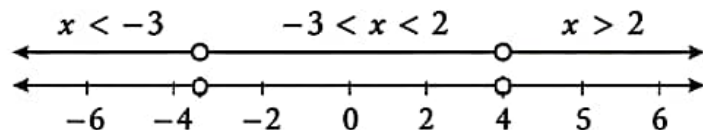
$$x = -3, 2$$

Step II Determine intervals.

Interval 1: $x < -3$

Interval 2: $-3 < x < 2$

Interval 3: $x > 2$



Step III Test points in each interval.

For $x = -4$ (in $x < -3$):

$$(-4)^2 + (-4) - 6 = 16 - 4 - 6 = 6 > 0 \quad \times$$

For $x = 0$ (in $-3 < x < 2$):

$$(0)^2 + (0) - 6 = -6 < 0 \quad \checkmark$$

For $x = 3$ (in $x > 2$):

$$(3)^2 + 3 - 6 = 9 + 3 - 6 = 6 > 0 \quad \times$$

Step IV Conclusion:

The solution of the inequality is

$$(-3, 2)$$

(iv) $x^2 - 6x + 9 < 0$

Step I Solve the corresponding equation.

$$x^2 - 6x + 9 = 0$$

$$(x - 3)^2 = 0$$

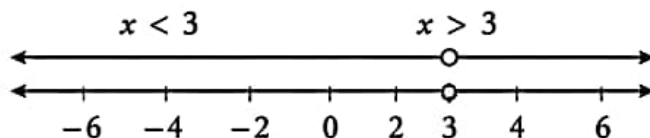
$$x - 3 = 0$$

$$x = 3$$

Step II Determine intervals.

Interval 1: $x < 3$

Interval 2: $x > 3$



Step III Test points in each interval.

For $x = 0$ (in $x < 3$):

$$0^2 - 6(0) + 9 = 9 > 0 \quad \times$$

For $x = 4$ (in $x > 3$):

$$4^2 - 6(4) + 9 = 16 - 24 + 9 = 1 > 0 \quad \times$$

Step IV Conclusion:

Since the quadratic expression is always greater than zero and never less than zero (at $x = 3$ it is equal to zero), the solution set is

$$\text{No solution } (\emptyset)$$

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QUESTION 1 Solve the following inequalities:

(v) $4x^2 - 16x + 15 \leq 0$

(vi) $-x^2 + 3x - 2 \geq 0$

SOLUTION:

(v) $4x^2 - 16x + 15 \leq 0$

Step I Solve the corresponding equation.

$$4x^2 - 16x + 15 = 0$$

$$4(x^2 - 4x) + 15 = 0$$

$$4(x^2 - 4x + 4 - 4) + 15 = 0$$

$$4(x - 2)^2 - 1 = 0$$

$$4(x - 2)^2 = 1$$

$$(x - 2)^2 = \frac{1}{4}$$

$$x - 2 = \pm \frac{1}{2}$$

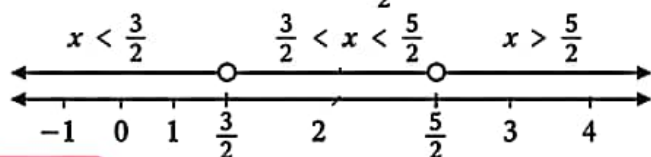
$$x = \frac{3}{2}, \frac{5}{2}$$

Step II Determine intervals.

Interval 1: $x < \frac{3}{2}$

Interval 2: $\frac{3}{2} < x < \frac{5}{2}$

Interval 3: $x > \frac{5}{2}$



Step III Test points in each interval.

For $x = 0$ (in $x < \frac{3}{2}$):

$$4(0)^2 - 16(0) + 15 = 15 > 0 \quad \times$$

For $x = 2$ (in $\frac{3}{2} < x < \frac{5}{2}$):

$$4(2)^2 - 16(2) + 15 = 16 - 32 + 15 = -1 < 0 \quad \checkmark$$

For $x = 3$ (in $x > \frac{5}{2}$):

$$4(3)^2 - 16(3) + 15 = 36 - 48 + 15 = 3 > 0 \quad \times$$

Step IV Conclusion:

The solution of the inequality is

$$\frac{3}{2} \leq x \leq \frac{5}{2}$$

(vi) $-x^2 + 3x - 2 \geq 0$

Step I Solve the corresponding equation.

$$-x^2 + 3x - 2 = 0$$

$$x^2 - 3x + 2 = 0$$

$$(x - 1)(x - 2) = 0$$

$$x - 1 = 0 \quad \text{or} \quad x - 2 = 0$$

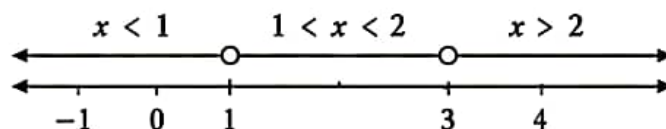
$$x = 1, 2$$

Step II Determine intervals.

Interval 1: $x < 1$

Interval 2: $1 < x < 2$

Interval 3: $x > 2$



Step III Test points in each interval.

For $x = 0$ (in $x < 1$):

$$-(0)^2 + 3(0) - 2 = -2 < 0 \quad \times$$

For $x = \frac{3}{2}$ (in $1 < x < 2$):

$$-\left(\frac{3}{2}\right)^2 + 3\left(\frac{3}{2}\right) - 2 = -\frac{9}{4} + \frac{9}{2} - 2 = \frac{1}{4} > 0 \quad \checkmark$$

For $x = 3$ (in $x > 2$):

$$-(3)^2 + 3(3) - 2 = -9 + 9 - 2 = -2 < 0 \quad \times$$

Step IV Conclusion:

The solution of the inequality is

$$1 \leq x \leq 2$$

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QUESTION 1 Make F the subject of the formula, $C = \frac{5}{9}(F - 32)$.

ANSWER:

$$\begin{aligned}C &= \frac{5}{9}(F - 32) \\ \Rightarrow \frac{9}{5}C &= F - 32 \\ \Rightarrow F &= \frac{9}{5}C + 32\end{aligned}$$

$$F = \frac{9}{5}C + 32$$

QUESTION 2 The formula for finding simple interest is $I = PRT$.

(a) Make P the subject of the formula. (b) Make T the subject of the formula.

ANSWER:

(a) P the subject:

$$\begin{aligned}I &= PRT \\ \Rightarrow \frac{I}{RT} &= P \\ P &= \frac{I}{RT}\end{aligned}$$

(b) T the subject:

$$\begin{aligned}I &= PRT \\ \Rightarrow \frac{I}{PR} &= T \\ T &= \frac{I}{PR}\end{aligned}$$

QUESTION 3 Make ' a ' the subject of the formula $S = 2a + (n - 1)d$.

ANSWER:

$$\begin{aligned}S &= 2a + (n - 1)d \\ \Rightarrow 2a &= S - (n - 1)d \\ \Rightarrow a &= \frac{S - (n - 1)d}{2}\end{aligned}$$

$$a = \frac{S - (n - 1)d}{2}$$

QUESTION 4 The volume of a cylinder is given by the formula, $V = \pi r^2 h$.
Make ' h ' the subject of the formula.

ANSWER:

$$\begin{aligned}V &= \pi r^2 h \\ \Rightarrow \frac{V}{\pi r^2} &= h\end{aligned}$$

$$h = \frac{V}{\pi r^2}$$

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QUESTION 5

The area of a trapezoid is $A = \frac{1}{2} h (b_1 + b_2)$, make h the subject of the formula.

ANSWER:

$$A = \frac{1}{2} h (b_1 + b_2)$$

$$\Rightarrow 2A = h(b_1 + b_2)$$

$$\Rightarrow h = \frac{2A}{b_1 + b_2}$$

$$h = \frac{2A}{b_1 + b_2}$$

QUESTION 6

If $y = mx + c$, then make 'x' the subject of this equation.

ANSWER:

$$y = mx + c$$

$$\Rightarrow y - c = mx$$

$$\Rightarrow x = \frac{y - c}{m}$$

$$x = \frac{y - c}{m}$$

QUESTION 7

Perimeter (P) of a rectangle is $P = 2(\ell + w)$, make ℓ as the subject of this formula.

ANSWER:

$$P = 2(\ell + w)$$

$$\Rightarrow \frac{P}{2} = \ell + w$$

$$\Rightarrow \ell = \frac{P}{2} - w$$

$$\ell = \frac{P}{2} - w$$

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QUESTION 8

The equation of a parabola is $y^2 = 4ax$, make 'x' as a subject of this equation.

ANSWER:

$$y^2 = 4ax$$

$$\Rightarrow 4ax = y^2$$

$$\Rightarrow x = \frac{y^2}{4a}$$

$$x = \frac{y^2}{4a}$$

QUESTION 9

If $P = S - C$, where S is selling price and C is cost price.

Make S as subject of the equation.

ANSWER:

$$P = S - C$$

$$\Rightarrow P + C = S$$

$$\Rightarrow S = P + C$$

$$S = P + C$$

QUESTION 10

Volume of the cone is $V = \frac{1}{3} \pi r^2 h$, make 'h' as subject of this formula.

ANSWER:

$$V = \frac{1}{3} \pi r^2 h$$

$$\Rightarrow 3V = \pi r^2 h$$

$$\Rightarrow h = \frac{3V}{\pi r^2}$$

$$h = \frac{3V}{\pi r^2}$$



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EXERCISE 2.7

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QUESTION 1

A town's population is modeled by $P(t) = -2t^2 + 40t + 800$, where t is years since 2020. Find the years when the population will be at least 1000.

ANSWER:

Population at least 1000 means,

$$P(t) \geq 1000$$

$$-2t^2 + 40t + 800 \geq 1000$$

$$\Rightarrow -2t^2 + 40t - 200 \geq 0$$

$$\Rightarrow t^2 - 20t + 100 \leq 0 \quad [\text{Multiplying by } -1]$$

$$\Rightarrow (t - 10)^2 \leq 0$$

$$\Rightarrow t - 10 = 0 \Rightarrow t = 10$$

$$\text{Since, } (t - 10)^2 \leq 0$$

is satisfied only when $t = 10$

$$t = 10$$

So, the population will be at least 1000 after 10 years from 2020.

$$\text{Year} = 2020 + 10 = 2030$$

Therefore, the population will be at least 1000 in the year 2030.

QUESTION 2

A company models its profit P in thousands of rupees by the equation: $P(x) = -5x^2 + 150x - 1000$, where x is the price per item in rupees. Find the price that gives maximum profit.

ANSWER:

The profit function is:

$$P(x) = -5x^2 + 150x - 1000$$

This is a quadratic function of the form

$$ax^2 + bx + c, \text{ where}$$

$$a = -5, \quad b = 150, \quad c = -1000$$

Since $a = -5 < 0$,

so the parabola opens downward and hence the function has maximum value.

The maximum profit occurs at:

$$x = -\frac{b}{2a}$$

$$\Rightarrow x = -\frac{150}{2(-5)}$$

$$\Rightarrow x = -\frac{150}{-10} = 15$$

Now, price for maximum profit:

$$x = 15 \text{ rupees}$$

Maximum profit:

$$P(15) = -5(15)^2 + 150(15) - 1000$$

$$= -5(225) + 2250 - 1000$$

$$= -1125 + 2250 - 1000$$

$$= 125 \text{ (in thousands of rupees)}$$

$$\text{Maximum profit} = 125 \text{ thousand rupees}$$

Therefore, the price that gives maximum profit is 15 rupees.

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QUESTION 3

A toy car rolls down an incline and covers a distance given by the equation $d = t^2 - 0.5t$ metres, where t is the time in seconds.

Find the time when the car has travelled a distance 12.5 metres.

ANSWER:

Given, $d = t^2 - 0.5t$

The car has travelled 12.5 metres.

So, $d = 12.5$

$$t^2 - 0.5t = 12.5$$

$$\Rightarrow t^2 - 0.5t - 12.5 = 0$$

$$\Rightarrow 2t^2 - t - 25 = 0 \quad [\text{Multiplying by 2}]$$

Using quadratic formula,

$$\begin{aligned} t &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{-(-1) \pm \sqrt{(-1)^2 - 4(2)(-25)}}{2(2)} \\ &= \frac{1 \pm \sqrt{1 + 200}}{4} \\ &= \frac{1 \pm \sqrt{201}}{4} \end{aligned}$$

So,

$$t = \frac{1 + \sqrt{201}}{4}, \quad t = \frac{1 - \sqrt{201}}{4}$$

Since time cannot be negative,
we take positive value.

$$t = \frac{1 + \sqrt{201}}{4} \text{ seconds}$$

Approximately,

$$t \approx 3.80 \text{ seconds}$$

QUESTION 4

A ball's height (in metres) after t seconds is $h(t) = -4t^2 + 24t$.

For what time interval is the ball at least 20m above the ground?

ANSWER:

At least 20m above the ground means,

$$h(t) \geq 20$$

$$-4t^2 + 24t \geq 20$$

$$\Rightarrow -4t^2 + 24t - 20 \geq 0$$

$$\Rightarrow -4(t^2 - 6t + 5) \geq 0$$

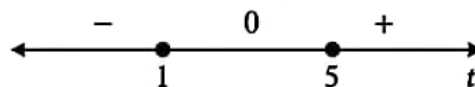
$$\Rightarrow t^2 - 6t + 5 \leq 0 \quad [\text{Dividing by } -4 \\ \text{(and reversing the sign)}]$$

$$\Rightarrow (t - 1)(t - 5) \leq 0$$

The quadratic $t^2 - 6t + 5$ has roots:

$$t - 1 = 0 \Rightarrow t = 1, \quad t - 5 = 0 \Rightarrow t = 5$$

Since the coefficient of t^2 is positive,
the expression $t^2 - 6t + 5$ is ≤ 0
between the roots (inclusive).



So,

$$1 \leq t \leq 5 \text{ seconds}$$

Hence, the ball is at least 20 metres
above the ground for

$$1 \leq t \leq 5 \text{ seconds.}$$

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QUESTION 5

A ball is thrown upward with an initial velocity of 40 ms^{-1} .

Calculate the maximum height it reaches above ground level.

ANSWER:

At maximum height, final velocity $v_f = 0$.

Given:

Initial velocity, $v_i = 40 \text{ ms}^{-1}$

Final velocity, $v_f = 0 \text{ ms}^{-1}$

Acceleration due to gravity, $g = -10 \text{ ms}^{-2}$

Maximum height, $s = ?$

Using equation of motion: $v_f^2 = v_i^2 + 2gs$

$$(0)^2 = (40)^2 + 2(-10)s$$

$$0 = 1600 - 20s$$

$$20s = 1600$$

$$s = \frac{1600}{20} = 80$$

Maximum height
= 80 metres

QUESTION 6

A freelancer's earnings follow the model $E(h) = -2h^2 + 40h$,
where E is earning in rupees and h is hours worked per week.

What is the maximum number of hours he should work to maximize earnings?

ANSWER:

The earning function is:

$$E(h) = -2h^2 + 40h$$

This is a quadratic function of the form

$ah^2 + bh + c$, where

$$a = -2, \quad b = 40, \quad c = 0$$

Since $a = -2 < 0$,

so the parabola opens downward and
hence the function has maximum value.

The maximum earning occurs at:

$$h = -\frac{b}{2a}$$

$$\Rightarrow h = -\frac{40}{2(-2)}$$

$$\Rightarrow h = -\frac{40}{-4} = 10$$

Therefore, the maximum number of
hours he should work is

$h = 10 \text{ hours per week}$

Maximum earning:

$$E(10) = -2(10)^2 + 40(10)$$

$$= -2(100) + 400$$

$$= -200 + 400$$

$$= 200 \text{ rupees}$$

Maximum earning = 200 rupees



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2. Solve the following quadratic equations by factorization method, by completing square method and by quadratic formula:

(i) $8x^2 = x + 7$

Solution:

Given equation is

$$8x^2 = x + 7$$

$$\Rightarrow 8x^2 - x - 7 = 0 \quad \text{--- (1)}$$

(a) By factorization method:

$$8x^2 - x - 7 = 0$$

$$\Rightarrow 8x^2 - 8x + 7x - 7 = 0$$

$$\Rightarrow 8x(x-1) + 7(x-1) = 0$$

$$\Rightarrow (8x+7)(x-1) = 0$$

$$\Rightarrow 8x+7 = 0 \quad \text{or} \quad x-1 = 0$$

$$\Rightarrow x = -\frac{7}{8} \quad \text{or} \quad x = 1$$

$\therefore$ The solutions are $x = 1, -\frac{7}{8}$

(b) By completing square method:

$$8x^2 - x - 7 = 0$$

$$\Rightarrow 8x^2 - x = 7$$

$$\Rightarrow x^2 - \frac{1}{8}x = \frac{7}{8}$$

$$\Rightarrow x^2 - \frac{1}{8}x + \left(\frac{1}{16}\right)^2 = \frac{7}{8} + \left(\frac{1}{16}\right)^2$$

$$\Rightarrow \left(x - \frac{1}{16}\right)^2 = \frac{113}{256}$$

$$\Rightarrow x - \frac{1}{16} = \pm \sqrt{\frac{113}{256}}$$

$$\Rightarrow x = \frac{1}{16} \pm \frac{\sqrt{113}}{16}$$

$$\Rightarrow x = \frac{1 \pm \sqrt{113}}{16}$$

(c) By quadratic formula:

For $8x^2 - x - 7 = 0$, $a = 8$, $b = -1$, $c = -7$

$$\begin{aligned} x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{-(-1) \pm \sqrt{(-1)^2 - 4(8)(-7)}}{2(8)} \\ &= \frac{1 \pm \sqrt{1 + 224}}{16} \\ &= \frac{1 \pm \sqrt{225}}{16} \\ &= \frac{1 \pm 15}{16} \end{aligned}$$

$$\Rightarrow x = \frac{1 + 15}{16} \quad \text{or} \quad x = \frac{1 - 15}{16}$$

$$\Rightarrow x = \frac{16}{16} \quad \text{or} \quad x = \frac{-14}{16}$$

$$\Rightarrow x = 1 \quad \text{or} \quad x = -\frac{7}{8}$$

$\therefore$ The solutions are $x = 1, -\frac{7}{8}$



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PREPARATION SERIES

2. Solve the following quadratic equations by factorization method, by completing square method and by quadratic formula:

(iii) $x^2 + 10x + 25 = 0$

Solution:

Given equation is

$$x^2 + 10x + 25 = 0 \quad \text{--- (1)}$$

(a) By factorization method:

$$x^2 + 10x + 25 = 0$$

$$\Rightarrow x^2 + 5x + 5x + 25 = 0$$

$$\Rightarrow x(x + 5) + 5(x + 5) = 0$$

$$\Rightarrow (x + 5)(x + 5) = 0$$

$$\Rightarrow x + 5 = 0 \quad \text{or} \quad x + 5 = 0$$

$$\Rightarrow x = -5 \quad \text{or} \quad x = -5$$

$\therefore$ The solutions are $x = -5, -5$

(b) By completing square method:

$$x^2 + 10x + 25 = 0$$

$$\Rightarrow x^2 + 10x = -25$$

$$\Rightarrow x^2 + 10x + \left(\frac{10}{2}\right)^2 = -25 + \left(\frac{10}{2}\right)^2$$

$$\Rightarrow (x + 5)^2 = -25 + 25$$

$$\Rightarrow (x + 5)^2 = 0$$

$$\Rightarrow x + 5 = \pm \sqrt{0}$$

$$\Rightarrow x + 5 = \pm 0$$

$$\Rightarrow x + 5 = 0 \quad \text{or} \quad x + 5 = 0$$

$$\Rightarrow x = -5 \quad \text{or} \quad x = -5$$

$\therefore$ The solutions are $x = -5, -5$

(c) By quadratic formula:

For $x^2 + 10x + 25 = 0$, $a = 1$, $b = 10$, $c = 25$

$$\begin{aligned} x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{-10 \pm \sqrt{(10)^2 - 4(1)(25)}}{2(1)} \\ &= \frac{-10 \pm \sqrt{100 - 100}}{2} \\ &= \frac{-10 \pm \sqrt{0}}{2} \\ &= \frac{-10 \pm 0}{2} \\ &= \frac{-10}{2} \end{aligned}$$

$$\Rightarrow x = \frac{-10}{2} \quad \text{or} \quad x = \frac{-10}{2}$$

$$\Rightarrow x = -5 \quad \text{or} \quad x = -5$$

$\therefore$ The solutions are $x = -5, -5$



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3. Form a quadratic equation whose roots are; $-4, \frac{5}{2}$.

Solution:

Let the roots of the required quadratic equation be $\alpha = -4$ and $\beta = \frac{5}{2}$.

$$\text{Sum of roots} = \alpha + \beta = -4 + \frac{5}{2} = \frac{-8 + 5}{2} = \frac{-3}{2}$$

$$\text{Product of roots} = \alpha\beta = -4 \times \frac{5}{2} = \frac{-20}{2} = -10$$

A quadratic equation whose sum of roots is $(\alpha + \beta)$ and product of roots is $(\alpha\beta)$ is given by $x^2 - (\alpha + \beta)x + \alpha\beta = 0$

$$\Rightarrow x^2 - \left(\frac{-3}{2}\right)x - 10 = 0 \quad [\text{Multiplying by 2 to clear the fraction}]$$

$$\Rightarrow 2x^2 + 3x - 20 = 0$$

$\therefore$ The required quadratic equation is $2x^2 + 3x - 20 = 0$.

4. Examine the nature of the roots of the following equations:

(i) $2x^2 + 7x + 3 = 0$

Solution:

(i) Here, $a = 2$, $b = 7$, $c = 3$

$$\text{Discriminant } D = b^2 - 4ac$$

$$D = (7)^2 - 4(2)(3)$$

$$D = 49 - 24$$

$$D = 25$$

Since $D > 0$ and is a perfect square,
the roots are **real, distinct and rational**.

(ii) $3x^2 + 4x + 5 = 0$

Solution:

(ii) Here, $a = 3$, $b = 4$, $c = 5$

$$\text{Discriminant } D = b^2 - 4ac$$

$$D = (4)^2 - 4(3)(5)$$

$$D = 16 - 60$$

$$D = -44$$

Since $D < 0$ and is not a perfect square,
the roots are **real, distinct and irrational**.

Summary:

If $\Delta > 0$ and is a perfect square $\Rightarrow$ Roots are real, distinct and rational.

If $\Delta > 0$ and is not a perfect square $\Rightarrow$ Roots are real, distinct and irrational.

If $\Delta = 0$ $\Rightarrow$ Roots are real and equal.

If $\Delta < 0$ $\Rightarrow$ Roots are imaginary.

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REVIEW EXERCISE 2

BOARD EXAM
PREPARATION SERIES

5. If a ball is thrown upward with a velocity v , the maximum height it reaches can be determined by using a formula $h = \frac{v^2}{2g}$. Rearrange the formula to make v the subject.

Solution:

$$h = \frac{v^2}{2g}$$

$$\Rightarrow 2gh = v^2$$

$$\Rightarrow v^2 = 2gh$$

$$\Rightarrow v = \pm \sqrt{2gh}$$

$\therefore$ The required formula with v as the subject is $v = \pm \sqrt{2gh}$.

6. If the equation $x^2 + 2(1+k)x + k^2 = 0$ has equal roots, then find the value of k .

Solution:

Given equation is $x^2 + 2(1+k)x + k^2 = 0$

Here, $a = 1$, $b = 2(1+k)$, $c = k^2$

For equal roots, Discriminant $D = 0$.

$$D = b^2 - 4ac$$

$$0 = [2(1+k)]^2 - 4(1)(k^2)$$

$$0 = 4(1+k)^2 - 4k^2$$

$$0 = 4(1 + 2k + k^2) - 4k^2$$

$$0 = 4 + 8k + 4k^2 - 4k^2$$

$$0 = 4 + 8k$$

$$8k = -4$$

$$k = -\frac{4}{8} = -\frac{1}{2}$$

$\therefore$ The value of k is $-\frac{1}{2}$.